



Soil Compaction and Growth of Woody Plants

T.T. Kozlowski

To cite this article: T.T. Kozlowski (1999) Soil Compaction and Growth of Woody Plants, Scandinavian Journal of Forest Research, 14:6, 596-619, DOI: [10.1080/02827589908540825](https://doi.org/10.1080/02827589908540825)

To link to this article: <https://doi.org/10.1080/02827589908540825>



Published online: 14 Jan 2008.



Submit your article to this journal [↗](#)



Article views: 2785



View related articles [↗](#)



Citing articles: 70 View citing articles [↗](#)

Soil Compaction and Growth of Woody Plants

T. T. KOZLOWSKI*

Department of Environmental Science, Policy and Management, College of Natural Resources, University of California, Berkeley, CA 94720, USA

Scandinavian Journal
of Forest Research



Kozlowski, T. T. (Department of Environmental Science, Policy and Management, College of Natural Resources, University of California, Berkeley, CA 94720, USA). *Soil compaction and growth of woody plants*. Received April 24, 1998. Accepted Oct. 18, 1998. Scand. J. For. Res. 14: 596–619, 1999.

Although soil compaction in the field may benefit or inhibit the growth of plants, the harmful effects are much more common. This paper emphasizes the deleterious effects of predominantly high levels of soil compaction on plant growth and yield. High levels of soil compaction are common in heavily used recreation areas, construction sites, urban areas, timber harvesting sites, fruit orchards, agroforestry systems and tree nurseries. Compaction can occur naturally by settling or slumping of soil or may be induced by tillage tools, heavy machinery, pedestrian traffic, trampling by animals and fire. Compaction typically alters soil structure and hydrology by increasing soil bulk density; breaking down soil aggregates; decreasing soil porosity, aeration and infiltration capacity; and by increasing soil strength, water runoff and soil erosion. Appreciable compaction of soil leads to physiological dysfunctions in plants. Often, but not always, reduced water absorption and leaf water deficits develop. Soil compaction also induces changes in the amounts and balances of growth hormones in plants, especially increases in abscisic acid and ethylene. Absorption of the major mineral nutrients is reduced by compaction of both surface soils and subsoils. The rate of photosynthesis of plants growing in very compacted soil is decreased by both stomatal and non-stomatal inhibition. Total photosynthesis is reduced as a result of smaller leaf areas. As soils become increasingly compacted respiration of roots shifts toward an anaerobic state. Severe soil compaction adversely influences regeneration of forest stands by inhibiting seed germination and growth of seedlings, and by inducing seedling mortality. Growth of woody plants beyond the seedling stage and yields of harvestable plant products also are greatly decreased by soil compaction because of the combined effects of high soil strength, decreased infiltration of water and poor soil aeration, all of which lead to a decreased supply of physiological growth requirements at meristematic sites. Many protocols have been developed, with variable success, to alleviate the adverse effects of soil compaction on the growth and development of woody plants. These include planting of compaction-tolerant species, controlling vehicular and animal traffic, amending soils by adding coarse materials and/or organic matter, replacing compacted soils with uncompacted soils, loosening soils with aerating equipment, installing drainage systems and judiciously applying fertilizers. Prevention of soil compaction before planting is usually much preferred over post-planting treatments because the latter are expensive and difficult to apply, may not be adequately effective and may injure plant roots. *Key words:* aeration, amelioration, bulk density, compaction, drainage, erosion, fire, hormones, hydrology, infiltration, mineral nutrition, mortality, mycorrhizae, photosynthesis, plant growth, respiration, runoff, soil strength, stand regeneration, water deficits.

INTRODUCTION

The effects of soil compaction on plant growth are very complex and vary from beneficial to harmful, with the latter being much more common. Mild compaction of soil after seeds are planted may benefit plant growth and development by improving capillary movement of water to the seeds. Nevertheless, a voluminous literature shows that soil compaction in

the field is predominantly severe enough to create conditions that lead to inhibition of plant growth and yield. Furthermore, there is growing concern that progressively severe levels of human-induced soil compaction will occur as the world population increases. With these considerations in mind this overview emphasizes the deleterious effects of relatively high levels of soil compaction on soil structure, hydrology and physiological processes that adversely affect the growth and yield of woody plants.

Soil compaction is a worldwide problem of enormous economic and ecological importance. In the

* Present address: 2855 Carlsbad Blvd, S-326 Carlsbad, CA 92008, USA.

USA alone it was estimated that in 1971 soil compaction accounted for an annual loss in crop values of \$1.2 billion (Gill 1971). Current losses undoubtedly are much greater. Growth inhibition and/or mortality of woody plants by soil compaction have been thoroughly documented in recreation areas such as parks and golf courses (Davis 1952, Lunt 1956, Madison 1971, Carrow 1981, Sills & Carrow 1983, Carrow & Petrovic 1992), construction sites (Alberty et al. 1984, Randrup 1997), urban areas (Patterson 1977, Yingling et al. 1979, Jim 1993), timber harvesting areas (Youngberg 1959, Sands & Bowen 1978, Cochran & Brock 1985, Reisinger et al. 1988, Stewart et al. 1988, Firth & Murphy 1989, Mohd. Basri & Nik Muhamad 1992), fruit orchards (Richards & Cockroft 1974), agroforestry systems (Wairiu et al. 1993) and tree nurseries (Boyer & South 1988).

The degree of soil compaction depends on soil texture, pH, cation-exchange capacity, clay particle thickness and water content, and on the presence of organic matter, iron oxides and free aluminum hydroxide, which determine the nature of the resulting cohesive forces between soil constituents (Assouline et al. 1997).

Surface pressure applied to soils causes breaking of bonds of aggregating agents that hold soil particles together into structural units. Subsequently, the particles become reoriented into a configuration that has a higher bulk density (BD) (mass per unit volume) (Cassel 1983). Clay soils and those with a high proportion of exchangeable Ca are readily compacted, especially when wet. Both dry and coarse-textured soils are less likely to become compacted (Snider & Miller 1985, Kozlowski et al. 1991). Dry soils resist compaction because of their stiff matrices, high degree of bonding of particles, interlocking and frictional resistance to deformation (Hillel 1982). Volcanic soils may compact and remain compacted. As an unsaturated soil becomes wetter, the moisture films decrease the interparticle bonds, apparently reducing friction and increasing compaction. However, when a soil approaches saturation, the proportional volume of air that can be expelled is decreased and a given externally applied pressure cannot compact the soil as much as it previously could. Hence, an additional increase in soil wetness lowers its compactibility (Hillel 1982).

Severely compacted soils typically do not recover naturally for a very long time. Subsoil compaction by heavy vehicles persisted for more than 4 years (Lowery & Schuler 1991). In Australian forests there was

little recovery of compacted soils for at least 7 years (Cheatle 1991) and soils of the boreal forest of Canada remained compacted for several decades (Corns 1988). In Californian forests surface soil compaction from tractor skidding may last for at least 40 years (Vora 1988). In the southern USA soil recovery from compaction is very slow because of the absence of freezing and thawing. Natural recovery of some southern soils requires up to 60 years (Mitchell et al. 1982). Recovery of compacted soils varies with the depth of the compacted layer. For example, recovery to the original compaction level of the surface soil (0–0.75 cm depth) of a sandy loam required 4.5–8.5 years; no recovery was evident after 8.5 years for 15–23 cm depths or 23–30 cm depths (Thorud & Frissell 1976).

CAUSES OF SOIL COMPACTION

Any force exerted on the soil may compact it. Soils compact both naturally and because of the activities of humans and trampling by animals. Loose soils may be compressed as a result of settling and/or slumping, especially when very wet (Koolen & Kuipers 1983). Settling typically is associated with recurrent drying and wetting, hence shrinkage and swelling of soil, that leads to closer packing of aggregates. Saturation of sandy topsoils during flood irrigation can cause enough settling of soil particles to increase compaction. Slumping follows weakening of soil aggregates as a result of wetting and partial disintegration at contact points, thus allowing for closer packing (Mullins 1991).

As tree roots grow they transmit the weights of trees on to the soil, hence compacting it. Severe forest fires also create conditions for soil compaction and development of water-repellent surface layers (Wright & Bailey 1982). Surface crusts may also form over exposed soils by the actions of rain and drying. Soil crusts typically consist of a thin surface skin about 0.1 mm thick and a layer of clogged pores just below the surface (Chen et al. 1980). The strengths of soil crusts increase with the soil moisture content at the time of compaction. When a dry soil becomes compacted the strength of the crust increases if a rain follows (Hegarty & Royle 1978).

Soils may be compacted by tillage tools that operate below the soil surface. As a tool is thrust into a soil some of the soil is pushed ahead of the moving tool against the resistance of the soil body; hence, the soil becomes compacted (Hillel 1982). In wet soils

tillage equipment often creates a dense tillage pan just below the loosened surface layer. However, such pans are often caused more by smearing than by compaction. Most commonly, soils are severely compacted by heavy machinery, pedestrian traffic and trampling by domestic and wild animals. Recovery of compacted soils depends on freezing and thawing, swelling and shrinking, activity of soil animals and especially earthworms, and the disruptive action of plant roots. Compacted layers are often found at different soil depths. Compaction may be shallow when caused by trampling, of medium depth when induced by vehicle traffic and deep because of the effects of vibration of constant heavy vehicle traffic (Rolf 1994).

Machinery and heavy equipment

For a long time attention was largely centered on compaction of surface soil layers. However, as vehicle traffic intensified and axle loads increased, it became apparent that subsoil compaction was also important (Hadas 1994). Heavy equipment often causes soil compaction which may persist for many years (Soane et al. 1981a, 1981b, 1982, Håkansson et al. 1987, McColl 1995). Under heavy traffic loads some soils become compacted to a depth of 1 m and sometimes more. However, the highest degree of compaction typically occurs in the top 30 cm of the soil profile, which normally contains most of the root mass (Wingate-Hill & Jakobson 1982). Vehicles exert three major compacting forces on soils: (1) the normal vertical force due to the dynamic load of the wheels, (2) shear stress caused by slippage of wheels, and (3) vibration of engines through the tires. Most soils become compacted during the first few passes of a vehicle; subsequent passes generally have little additional effect but the impact varies with the load and soil strength (Lockaby & Vidrine 1984, Shetron et al. 1988). However, the deeper layers of many soils compact further after a few passes. The BD of the upper layer of a forest soil reached a maximum after four passes with a rubber-tired skidder and remained statistically constant after additional passes (Lenhard 1986). Smith (1987) developed a model for predicting the BD of soil at various depths as caused by different types of vehicles. Deschênes (1989) published a useful table listing total weights, tire or track dimensions and pressures exerted on soils by more than 100 types of machines used in timber harvesting.

Compaction of soils by logging equipment such as tractors and skidders and by dragging of logs is well documented. The extent of compaction varies with soil texture, soil moisture content, ground pressure and vibration of vehicles (Greacen & Sands 1980, Ole-Meiludie & Njau 1989, Jim 1993).

Compared with soil BD on trails not used for skidding of logs the BD in the 0–8 cm depth after logging averaged 41–52% higher on skid trails. Even 8 years later the soil BD at 0–30 cm soil depths on skid trails exceeded the BD of soil beyond the skid trails by 20% (Miller et al. 1996). Soil BD was greater after logging (range of 0.34–1.13 g cm⁻³) than before logging (range of 0.10–0.95 g cm⁻³) (Purser & Cundy 1992). In New Hampshire soil compaction during logging increased BD by 5–15% in the upper 20 cm of soil (Johnson et al. 1991). In a Douglas-fir forest logging machinery increased the BD of near surface soil by 20–25% (McNeel & Ballard 1992). In Chile logging machinery increased compaction and BD of a clay soil (Gayoso & Iroume 1991). In Malaysia logging machines increased soil BD and decreased both the total pore space and the water-holding capacity of the soil (Kamaruzaman 1991, Kamaruzaman & Nik Muhamad 1992a, 1992b).

Pedestrian traffic

Intensive recreational use of parks and forests increases soil compaction, destroys ground cover, decreases infiltration of water, and increases runoff and soil erosion. In the Washington DC area compaction of several soils by pedestrian traffic increased the soil BD from a range of 1.20–1.60 g cm⁻³ to a range of 1.70–2.20 g cm⁻³ (Patterson 1976). At a Michigan campsite the soil BD increased by 21% (Legg & Schneider 1977). On campsites in a wilderness area the soil BD typically was 15% higher and infiltration rates were 30% lower than in an adjacent forest. However, the BD in a heavily used site was no higher than on a lightly used site (Cole & Fichtler 1983). In Illinois soil compaction increased with camping use (Young & Gilmore 1976). In the Rocky Mountain National Park campgrounds the soil from heavily used sites had an average BD 55.3% higher than soil from lightly used sites. The compacting effect of intensive use was more pronounced in the 0–5 cm soil layer on lightly and moderately used sites, and at a depth of 13 cm or more on heavily used sites (Dotzenko et al. 1967).

Table 1. *Studies showing effects of soil compaction on soil structure and hydrology*

(a) Increase in soil bulk density (BD)	
Alberty et al. (1984)	McNeel & Ballard (1992)
Bezkorowajnyi et al. (1993)	Miller et al. (1996)
Carter (1990)	Minore et al. (1969)
Cassel (1983)	Misra & Gibbons (1996)
Conlin & van den Driessche (1996)	Monti & Mackintosh (1979)
Davis (1992)	Mulholland & Fullen (1991)
Demko (1994)	Patterson (1976, 1977)
Dickerson (1976)	Patterson & Mader (1982)
Dotzenko et al. (1967)	
Froehlich & McNabb (1984)	Plass & Van der Bruijnzeel (1993)
Gameda et al. (1994)	Purser & Cundy (1992)
Gayoso & Iroume (1991)	Rab (1994)
Gilman et al. (1987)	Randrup (1997)
Greacen & Sands (1980)	Rolf (1991)
Hajabbasi et al. (1997)	Sharda (1977)
Helms & Hipkin (1986)	Sheriff & Nambiar (1995)
Horn et al. (1995)	Shetron et al. (1988)
Howard (1964)	Short et al. (1986)
Huang et al. (1996)	Simmons & Pope (1987)
Jim (1993)	Smith (1987)
Johnson et al. (1991)	Stirzaker et al. (1996)
Kamaruzaman (1991)	Taylor et al. (1966)
Kamaruzaman & Nik Muhamad (1992a), Kamaruzaman & Nik Muhamad (1992b)	Thorud & Frissell (1976)
Knapp (1989)	Tuttle et al. (1988)
Krenzer et al. (1989)	Usman (1994)
Lachno et al. (1982)	Vepraskas (1994)
Legg & Schneider (1977)	Willatt & Pullar (1983)
Lenhard (1986)	Wolkowski (1990)
Lockaby & Vidrine (1984)	Woodward (1996)
Lowery & Schuler (1991)	Zisa et al. (1980)
(b) Decrease in macropores and total porosity	
Day & Bassuk (1993)	Monti & Mackintosh (1979)
Day et al. (1995)	Richards & Cockroft (1974)
Federer et al. (1961)	Ruark et al. (1982)
Frede (1987)	Tuttle et al. (1988)
Greacen & Sands (1980)	Vepraskas (1994)
Jusoff (1989)	Willatt & Pullar (1983)

Table 1. (Continued)

Kamaruzaman (1991)	Wolkowski (1990)
(c) Decrease in soil aeration	
Boll (1987)	Hansen & Bakken (1993)
Cannell (1977)	
Currie (1984)	Horn et al. (1995)
Fedotov (1989)	McAfee et al. (1989)
Frede (1987)	Ruark et al. (1983)
Gill & Mittler (1956)	Stepniewski et al. (1994)
Gilman et al. (1987)	Tackett & Pearson (1964)
Grable (1971)	Wolkowski (1990)
Greacen & Sands (1980)	
(d) Decrease in infiltration capacity	
Arend (1941)	Ole-Meiludie & Njau (1989)
Cassel (1983)	Plass & Van der Bruijnzeel (1993)
Cole (1983)	Purser & Cundy (1992)
Cole & Fichtler (1983)	Settergren & Cole (1970)
Cullen et al. (1991)	Sharda (1977)
Ferrero (1991)	Sojka et al. (1993)
Gifford & Hawkins (1978)	Steinbrenner (1955)
Hatchell et al. (1970)	Steinbrenner & Gessell (1955)
Huang et al. (1996)	Tuttle et al. (1988)
Joshi et al. (1987)	Usman (1994)
Jusoff (1989)	Warren et al. (1987)
Kays & Patterson (1982)	van der Weert (1974)
Lutz (1945)	Wood et al. (1989)
Malmer & Grip (1990)	Zander et al. (1988)
Nimmo & Akstin (1988)	
(e) Increase in water runoff and soil erosion	
Anderson & Spencer (1991)	Nortcliff et al. (1990)
Edwards & Burney (1989, 1991)	Pinard et al. (1996)
Haig (1938)	Roth & Joschko (1991)
Humphreys & Craig (1981)	Wells et al. (1979)
Jim (1987a,b)	Wood et al. (1989)
Kang & Lal (1981)	Wright & Bailey (1982)
Kramer & Boyer (1995)	Zander et al. (1988)

Animal traffic

Trampling of soil by domestic and wild animals often severely compacts soils. Animal traffic is important because it generally occurs shortly after a rain when soils are very plastic (Tanner & Mamaril 1959, Howard 1964). The effects of animal traffic on compaction vary

with soil type, soil water content, seasonal climatic conditions and vegetation type (Warren et al. 1987).

In Wisconsin a variety of soils was compressed by animal traffic. Only a Warse silt loam, which contained little clay and acted mechanically like a very fine sand, did not compact significantly under animal traffic (Tanner & Mamaril 1959). Compaction of a silt loam in Wisconsin by animal traffic extended to a depth of at least 20 cm (Federer et al. 1961). Trampling of a dry soil by livestock in Texas compacted the surface layer and disrupted soil aggregation. Trampling of moist soil deformed the aggregates and created a flat, relatively impermeable layer comprised of dense and unstable clods (Warren et al. 1987). In the Mojave Desert trampling by sheep tripled the soil strength at a 12-cm depth (Webb & Stielstra 1979). In Scotland, even when the soil moisture content was near field capacity soil compaction was greater under trees than between trees. The differences were attributed to treading by sheep near the trees (Wairiu et al. 1993). Grazed pastures in Australia showed increases in soil BD with increased rates of animal stocking (Willatt & Pullar 1983). In New Zealand herds of introduced mammals compacted extensive areas of forest soils (Howard 1964).

Effects of compaction on soil structure and hydrology. Soil compaction is associated with several changes in soil structure and hydrology, especially: (1) increase in soil BD; (2) breakdown of soil aggregates; (3) decrease in soil porosity; (4) loss of pore continuity; (5) increase in shear strength; (6) decrease in infiltration capacity; and (7) increase in runoff of water and soil erosion (Jim 1993, Kozłowski & Pallardy 1997b).

Bulk density. A voluminous literature shows that compaction increases soil BD (mass per unit volume) (Table 1a). Such increases are induced by natural causes (Koolen & Kuipers 1983), vehicle traffic (Cassel 1983, Lenhard 1986, Smith 1987, Gayoso & Iroume 1991, Johnson et al. 1991, Kamaruzaman 1991, McNeel & Ballard 1992, Purser & Cundy 1992, Huang et al. 1996, Miller et al. 1996), pedestrian traffic (Dotzenko et al. 1967, Patterson 1976, Jim 1993) and trampling by animals (Howard 1964, Willatt & Pullar 1983, Bezkorowajnyi et al. 1993).

Soil porosity. Compaction decreases total pore volume in soil, especially the intercrumb pore volume, the most effective pore component influencing gas diffusion (Currie 1984; Table 1b). During soil compaction the proportion of macropore space is decreased and the proportion of micropore space is increased (Willatt & Pullar 1983, Huang et al. 1996).

Measurements along a gradient of increasing recreational use in the Boreal Forest region of Ontario, Canada, indicated that progressive soil compaction and loss of surface leaf litter were followed by formation of a thin-compacted mineral layer. Macropores in this layer were reduced up to 60%. Pores exceeding 2000 μm in size were essentially eliminated (Monti & Mackintosh 1979). Total pore space decreased as the soil was compacted in a forested recreational area in Malaysia. Average total porosity and pore volume of the surface soil decreased from 63% in an unused area to 52% in an area used for recreation (Jusoff 1989). The impact of water drops during irrigation of nursery beds may destroy soil aggregates, resulting in filtering of fine particles into macropores, thus reducing soil porosity (Tuttle et al. 1988).

Earthworms increase soil porosity by burrowing (Ehlers et al. 1983). Estimates of increases in soil porosity by earthworm activity range from negligible amounts (Springett et al. 1992) to as much as 100% (Edwards & Bohlen 1996). The effects of earthworms on porosity are mediated largely by an increase in soil macropores (Edwards & Bohlen 1992).

Soil aeration. Although gas movement in soil occurs primarily by diffusion it can also occur by mass flow, with the entire gas mixture moving in response to a pressure gradient. The rate of diffusion is proportional to porosity and inversely proportional to tortuosity (a measure of the interconnectedness of the pores). Mass flow is important only when pressure differences occur because of changes in soil water content, temperature or barometric pressure (Ball & Smith 1991).

The soil O_2 that is consumed by root respiration normally is replaced by diffusion from the above-ground atmosphere. Such gas exchange is impeded by soil compaction and by impermeable layers such as hardpans and pavements (Kozłowski & Pallardy 1997b). Soil aeration depends largely on the large pores that drain rapidly after rain or irrigation. Compaction of soil and pulverizing of soil aggregates often destroy enough of the large pores to impede O_2 transfer to plant roots and microorganisms (Table 1c). Aeration may not be a problem in coarse soils.

One of the most common causes of O_2 deficiency to roots is the placement of soil around the bases of tree stems. Yelenosky (1963) found the poorest soil aeration in an area where clay fill had been placed around tree stems. O_2 concentrations were as low as 1% and CO_2 content exceeded 20% under the fill. The O_2 content was lower and CO_2 content higher during

the growing season than during the dormant season. The next most poorly aerated soil was found where a driveway was under construction. The gaseous composition of the atmosphere changed drastically when layers of clay and gravel were laid down, rolled and packed. Within 2 weeks the soil O_2 content dropped from 20 to 4% and the CO_2 content increased from 0 to 16%. The O_2 content increased to 10% and CO_2 content decreased to 9.5% within 1 day after the clay-gravel layer was disturbed by a road grader. By comparison, in an undisturbed forest that was bordered by an unpaved parking lot the soil O_2 content was never lower than 18% and the CO_2 content never higher than 2%.

Infiltration of water. As the soil BD increases (and pore space decreases accordingly) the rate of infiltration of rain and irrigation water declines. Slow infiltration leads to soil water deficits, surface runoff, sheet erosion and gully formation (Cole 1982, 1986, Jim 1987a, 1987b, Wood et al. 1989, Kramer & Boyer 1995). In addition to surface-compacted layers, natural subsurface horizons (fragipans) with lower BD values than the soil above them inhibit both water and air movement (Cassel 1983).

There are many examples of reduced rates of water infiltration into severely compacted soils of the temperate zone and tropics (Table 1d). In western Montana water retention at a 15-cm soil depth was lower in areas with timber harvesting traffic than in areas without such traffic. Infiltration in severely trafficked areas was 81% less in volcanic ash over limestone till, 79% less in ash over quartzite till and 87% less in tertiary volcanic soil (Cullen et al. 1991). Water infiltration rates decreased by 25% in areas of light to moderate grazing and nearly 50% where grazing was heavy (Gifford & Hawkins 1978). In Connecticut State Parks the permeability of soil exposed to heavy recreational use was only a fraction of the permeability of undisturbed soil. The rate of infiltration varied appreciably with soil type. Infiltration into trampled Merrimac soil was approximately one-sixth of the rate in non-trampled areas. The effect of trampling of Holyoke sandy loam was even greater. Soil in the intensively used area was approximately one-twentieth as permeable as the soil in an unused area (Lutz 1945).

Adverse effects of severe soil compaction on infiltration of water in tropical soils have also been reported. In Sabah, Malaysia, soil compaction by tractors drastically lowered the water infiltration capacity, with the low rates persisting for up to 9 years

(Malmer & Grip 1990). In another study in Malaysian lowland rainforests the BD of the topsoil (0–46 cm depth) was highly correlated with steady-state infiltration rates. Average values were 88, 73 and 15 mm h^{-1} for undisturbed forest, recovering forest and tractor tracks, respectively (Plass & Van der Bruijnzeel 1993). In Tanzania soil compaction by logging practices substantially reduced the infiltration of water (Ole-Meiludie & Njau 1989). The reduced depth of infiltration in soil with increase in BD was attributed to fewer large pores, resulting in reduced availability of cross-sectional area for the flow of water (Sharda 1977).

Infiltration of water in compacted soil is often increased by earthworm burrows. Water infiltration was 4–10 times faster in soils containing earthworms than in soils without them (Edwards & Bohlen 1996). Elimination of earthworms from a grassland by pesticides decreased the water infiltration rate by 93% (Clements 1982).

Water runoff and soil erosion. A serious consequence of low permeability of severely compacted soils is runoff of water, which reduces soil water recharge and often leads to soil erosion and accelerated leaching of mineral nutrients (Kang & Lal 1981; Table 1e). Soil compaction is often accompanied by the development of a soil crust that reduces infiltration of water, hence increasing surface runoff (Malmer & Grip 1990). The formation of soil crusts often involves mechanical forces of rain splash, pedestrian traffic and/or the load of heavy machinery.

Surface runoff of water and erosion of topsoil were promoted by soil compaction (Anderson & Spencer 1991). In a tropical rainforest 16% of the rain went into runoff from logged plots as against 6% from unlogged forest (Nortcliff et al. 1990). In Bavaria surface runoff of water was rare in undisturbed forest but common on skid trails, even when there was little rain (Zander et al. 1988).

Forest fires often create conditions conducive to water runoff. Whereas high-intensity wildfires often lead to surface runoff, low-intensity fires do not (Humphreys & Craig 1981). If the forest floor is completely consumed by fire and the underlying soil exposed, the pores tend to become clogged and macropore space, infiltration of water and soil aeration decrease (Wells et al. 1979). Runoff in burned pine forests in the Sierras was 2–239 times as great as in unburned areas (Haig 1938). Decreases in soil moisture content following fire have been reported for many different sites (Ahlgren & Ahlgren 1960).

By consuming vegetation and decreasing infiltration capacity (Arend 1941) forest fires increase both sheet and rill erosion (Wright & Bailey 1982). Much erosion after fire is associated with water-repellent layers that form when soil particles are coated with hydrophobic organic compounds which are liberated during burning of litter. Such compounds move downwards into the soil and condense on the cooler particles (De Bano 1981). Where runoff cannot occur, as in depressions and ruts created by vehicle tires following logging of low-lying sites, poor drainage often creates locally waterlogged areas (Jim 1993).

PHYSIOLOGICAL RESPONSES OF PLANTS

Severe soil compaction typically leads to physiological dysfunctions in plants that alter availability of water, mineral nutrients, growth hormones and carbohydrates to meristematic sites as a prelude to reducing plant growth.

Water relations

Water deficits often, but not always, develop in shoots of plants growing in compacted soils. Water in the soil that is not penetrated by roots is largely unavailable to plants (Kozłowski 1972*b*, 1976, 1982*a*). Capillary movement of water from wet to dry portions of the soil is very slow. Because the soil immediately surrounding absorbing roots dries rapidly, water uptake by non-elongating roots is reduced, eventually leading to development of leaf water deficits. Hence, absorption of water by plants often decreases with increasing levels of soil compaction (Arvidsson & Jokela 1995, Sheriff & Nambiar 1995). Restriction of root growth by the high strength of severely compacted soils may also be expected to lead to plant water deficits.

Soil water deficits in compacted soil contributed to the development of water deficits in *Pinus radiata*. Appreciable subsoil compaction commenced between 30 and 40 cm below the soil surface. Needle water potential was lowered in plants growing in compacted soils. When the compacted layers were perforated the roots bypassed the compacted sections and extracted water from deeper soil depths, largely alleviating root water deficits (Nambiar & Sands 1992, Sheriff & Nambiar 1995).

The reduction of water uptake by roots of plants growing in compacted soil is not always followed by creation of appreciable shoot water deficits because stomatal closure by hormonal root signals may de-

crease transpirational water loss. For example, soil compaction decreased stomatal conductance in *Acacia confusa* and *Litsea glutinosa* but had little effect on leaf water potential (Liang et al. 1996). Stomatal closure was closely correlated with increases in abscisic acid (ABA) in the xylem of plants growing in compacted soils (Tardieu et al. 1991, 1992, Tardieu 1994, Mulholland et al. 1995, 1996).

When runoff of water does not occur and surface water accumulates in depressions and wheel ruts on compacted soils, leaf water deficits are not likely to develop. Several studies showed that stomata closed following soil flooding and without changes in leaf water potential (Pereira & Kozłowski 1977, Tang & Kozłowski 1982, Crane & Davies 1989, Pezeshki & Chambers 1985). Stomatal closure in flooded plants has been attributed to a hormonal signal transmitted from the roots to the shoots (Zhang & Davies 1990, Else et al. 1996).

Hormonal growth regulators

Soil compaction is often accompanied by changes in the amounts and balances of hormonal growth regulators in plants, primarily by substantial increases in ABA and ethylene. Severe soil compaction induced a rapid increase in xylem ABA (Tardieu et al. 1992, Tardieu 1994, Liang et al. 1996). The increase in ABA when soil BD is high is often correlated with a reduction in stomatal conductance, suggesting the involvement of ABA in root-to-shoot signalling and mediation of the effects of compaction stress (Mulholland et al. 1996).

As growing roots encounter mechanical impedance in soils, plants typically respond by producing ethylene above the basal level (Osborne 1976). Such stress-induced ethylene resulted from *de novo* synthesis rather than from facilitated diffusion of the ethylene already present in plant tissues (Sarquis et al. 1992).

When growth of *Vicia faba* roots was impeded by a barrier, the rate of ethylene evolution increased up to 6 times. When the barrier was removed the rate of ethylene production decreased to approximately the same rate as in unimpeded plants (Kays et al. 1974). Applied pressure stimulated ethylene evolution in *Zea mays* roots. After 1 h the rates of ethylene production increased by 10, 32, 70 and 255% at 25, 50, 75 and 100 kPa, respectively, and they continued to increase for at least 10 h. When seedlings were exposed to alternations of hourly cycles of pressure and relaxation of pressure, ethylene production rapidly in-

creased and decreased accordingly. Seedlings exposed to 1 μl of ethylene l^{-1} exhibited symptoms resembling those of plants grown under mechanical impedance (e.g. decreased root elongation and increased radial expansion of roots) (Sarquis et al. 1991). Under mechanical impedance both roots and shoots of *Z. mays* accumulated the ethylene precursor ACC (1-aminocyclopane-carboxylic acid). The concentration of ACC in roots more than doubled during the first hour of treatment, whereas in shoots it increased by 46%.

In vivo ACC synthase activity in roots was 7 times higher after the first hour of mechanical impedance. After 8 h the *in vivo* activity of the ethylene-forming enzyme (EFE) complex increased from 50 to 100%, depending on the amount of pressure applied. The data indicated that regulation of ethylene production under mechanical impedance was mediated by the action of ACC synthase, the EFE complex and ACC *N*-malonyltransferase, a conjugated form of ACC (Sarquis et al. 1992). In *Z. mays* seedlings hypoxia and mechanical impedance were strongly synergistic in their effects on ethylene evolution and ACC synthase activity. Such synergistic effects, as well as earlier plant responses to mechanical impedance than to hypoxia, suggested that different mechanisms were involved in the promotive effects of each stress on ethylene synthesis (He et al. 1996).

Mineral nutrition

Severe soil compaction typically decreases absorption by roots of major mineral nutrients, especially N, P and K. Mineral uptake is reduced by a combination of nutrient losses from the soil, decreased access of roots to nutrients and a lowered capacity of roots for mineral uptake (Kang & Lal 1981, Cole 1982, 1986, Silberbush et al. 1983, Bowen 1984, Allamaras et al. 1988, Zander et al. 1988, Arah & Smith 1989, Wolkowski 1990, Barber 1995, Kozłowski & Pallardy 1997a).

The concentrations of mineral nutrients decreased in *Pinus contorta* shoots as soil compaction increased (Conlin & van den Driessche 1996). Restricted nutrient uptake from compacted soil reduced nutrient concentrations in shoots of many species of forest trees (Comerford et al. 1984). Soil compaction decreased P uptake by *Eucalyptus* roots (Misra & Gibbons 1996). Application of a 90-kPa stress to a soil after planting reduced Ca, Mg and Mn uptake by herbaceous plants (Castilo et al. 1982). In Minnesota subsoil compaction reduced P and K uptake by up to

22% during years when June and July precipitation was below average (Dolan et al. 1992).

Soil compaction shifts soil conditions toward a hypoxic state that is associated with reduced aerobic microbial activity and increased denitrification. As soil compaction increases, the progressive reduction of macropores promotes microsite anaerobiosis (Torbert & Wood 1992). Gaseous losses of N typically were largest when soil compaction was greatest (Douglas & Crawford 1993). Denitrification was greater in untilled than in tilled soils. In untilled soils more anaerobic organisms and greater denitrification potential indicated less aerobic conditions than in tilled soils (Linn & Doran 1984). Broder et al. (1984) found higher populations of nitrifiers in the surface 15 cm of plowed soils than of unplowed soils.

Losses of N are caused largely by the action of denitrifying bacteria that reduce nitrates to molecular N (Kozłowski & Pallardy 1997a). Denitrification is characterized by coupling of nitrogen oxide reduction to electron transport phosphorylation, conversion of 80% or more of NO_3^- and NO_2^- nitrogen to N_2O and N_2 , rapid nitrate reduction and the presence of denitrifying types of nitrate reductases (Tiedje 1988). Whole soil denitrification rates are affected by the amount of soil hypoxia (Powlson 1993), soil texture (Groffman & Tiedje 1989, Nommik & Larsson 1989, Groffman et al. 1992), soil structure (Wolkowski 1990, Powlson 1993), moisture tension (Wolkowski 1990) and nitrate concentration (Vermes & Myrold 1992).

As soil particles are moved closer together during progressive soil compaction the rate of diffusion of mineral nutrients is accelerated. Whereas this may increase absorption of nutrients per unit length of root, nutrient uptake per plant is reduced because the growth of roots is inhibited by greater soil resistance to root penetration. When roots cannot explore a large volume of soil, mineral nutrients become positionally unavailable (Wolkowski 1990).

Mineral uptake involves the active transport of ions, which depends on expenditure of energy. Hence, absorption of mineral nutrients is influenced by both aeration and temperature, which affect plant metabolism (Kozłowski & Pallardy 1997a). Lowering the O_2 level in soil solutions from 90 to 50% equilibrium saturation with air decreased uptake of P, K and Ca by roots of *Pinus elliottii* (Shoulders & Ralston 1975). Active uptake of K by *P. elliottii* roots depended on the transport of O_2 from the upper parts of roots and/or stems exposed to air. Roots absorbed K when they were in an aerobic environment. Ab-

sorption of K stopped when N₂ replaced air in enclosures surrounding the lower stem and basal roots. Active K uptake responded directly to the level of available O₂ (Fisher & Stone 1990a, 1990b). At solution O₂ concentrations less than 1%, K leaked from the roots of *Prunus* trees but K uptake resumed when the soil was re-aerated (Rosen & Carlson 1984).

Mycorrhizae stimulate the growth of plants by increasing the absorption of poorly mobilized mineral nutrients (Simmons & Pope 1988a, 1988b, MacFall et al. 1991, O'Keefe & Sylvia 1991, Kozłowski & Pallardy 1997a). Reduced absorption of minerals by plants growing in compacted soil is partly traceable to poor mycorrhizal development. Mycorrhizal fungi that increased the growth of tree seedlings also increased their uptake of mineral nutrients (Burgess et al. 1993, Zabowski et al. 1994). Entry et al. (1996) demonstrated that the growth of mycorrhizal fungi, which are highly aerobic, was suppressed in compacted soils. The importance of soil aeration to mycorrhizal development is shown by the rare occurrence of mycorrhizae in flooded soils (Wilde 1954, Shuja et al. 1971, Filer 1975, Lodge 1986, Danell 1994).

Photosynthesis

Soil compaction typically reduces the rate of photosynthesis of plants as diverse as *Rubus* sp. (Wieniarska et al. 1987) and *Pinus contorta* (Conlin & van den Driessche 1996). The rate of photosynthesis is reduced when stomatal closure is associated with leaf water deficits (Kozłowski, 1972a) as well as when stomata close without a significant change in leaf water potential. Total photosynthesis of plants is reduced as a result of a smaller leaf area, reflecting both inhibition of leaf growth and increased leaf abscission (Kozłowski & Pallardy 1997a).

Reduction in the rate of photosynthesis of plants growing in compacted soils has been attributed to both stomatal and non-stomatal inhibition. An increase in the mechanical resistance of the soil is often followed by stomatal closure (Masle & Passioura 1987), which decreases photosynthesis by inhibiting the diffusion of CO₂ to the mesophyll. Correlations between stomatal aperture and photosynthetic rates were reported by Regehr et al. (1975) and Lakso (1979). Midday decreases in photosynthesis are often correlated with temporary stomatal closure (Tenhunen et al. 1982).

Non-stomatal inhibition of photosynthesis during a period of water deficit may involve decreases in carboxylating enzymes, capacity for electron transfer

and chlorophyll content (Kozłowski & Pallardy 1997a). However, the component processes of non-stomatal inhibition may vary appreciably. Photophosphorylation and photosynthetic carbon metabolism appear to be more sensitive than electron transfer to leaf dehydration (Epron et al. 1995, Kramer & Boyer 1995).

In a number of species reduced photosynthesis during brief periods of soil water shortage was attributed to both stomatal and non-stomatal inhibition. Examples are *Quercus alba*, *Q. stellata*, *Acer saccharum*, *Juglans nigra* (Ni & Pallardy 1992) and *Pinus taeda* (Teskey et al. 1986). The relative importance of stomatal vs non-stomatal inhibition under water stress varies with the drought tolerance of a species. For example, relative stomatal inhibition during drought increased in the drought-tolerant *Q. stellata* but decreased in the less drought-tolerant *A. saccharum* (Ni & Pallardy 1992). Stomatal inhibition of photosynthesis during water shortage was less important in wet-site species than in xeric species (Kubiske & Abrams 1993).

Photosynthesis of plants growing in compacted soil is often related to the availability of mineral nutrients. The inhibitory effects of mineral deficiency may reflect decreased chlorophyll synthesis, a lowered capacity for photosynthetic electron transport, decreased activity of carboxylating enzymes and stomatal closure. The total photosynthesis of mineral deficient plants is often reduced by a decrease in leaf area (Kozłowski & Pallardy 1997a).

Respiration

Soil compaction shifts soil conditions towards an anaerobic state. In most compacted soils both aerobic and some anaerobic respiration occur in roots. Where soils become severely compacted, and runoff of water is prevented and surface water accumulates, as in wheel ruts and depressions, much root respiration is largely anaerobic.

Aerobic root respiration is necessary for mineral uptake, synthesis of protoplasm and maintenance of cell membranes. In the absence of a sufficient supply of free O₂, the anaerobic respiration of roots is inefficient and does not release enough energy to maintain essential root functions, particularly the uptake of minerals (Kozłowski & Pallardy 1997a). A deficiency of soil O₂ also may adversely affect the membranes of root cells and cause loss of ions by leaching.

STAND REGENERATION AND TREE GROWTH

Severe compaction of soil adversely affects the growth of plants by a combination of physical soil changes and plant physiological dysfunctions (Jim 1993, Wronski & Murphy 1994). In dry soils increases in mechanical impedance and decreases in the soil water potential (and root pressure) predominate. In wet soils, however, O₂ deficiency and accumulating phytotoxins appear to be more important (Marschner 1995).

In early studies it was assumed that the rate of root elongation was reduced directly by the inhibition of cell elongation by external pressure. However, two lines of evidence subsequently emphasized that the major stimulus for responses of roots to small external pressures was not the physical restraint of cell enlargement: (1) the volume of cells was not always reduced by soil compaction and could even be increased as cell shapes were altered; and (2) there was a significant lag response, often of several days, between the relief of external pressure and the rate of root elongation. This lag suggested that internal physiological factors, presumably hormonal in nature, regulated the growth changes of plants (Goss & Russell 1980).

Roots in severely compacted soils appear to transmit more than one type of signal to the shoots. There appears to be a message that responds directly to soil mechanical impedance. This message affects growth largely during the first part of a growing cycle. It may be less important after roots have crossed a compacted layer. Imposed on this early signal are messages induced by other stresses caused by soil compaction (e.g. water stress and hypoxia). An example is the increase in xylem ABA of plants. In addition, changes in events that alter carbohydrate availability (e.g. reduced photosynthesis per unit of leaf area, smaller leaf area and changes in source-sink relations) modify the growth of root systems (Tardieu 1994). Superimposed on these changes are the inhibitory effects of mineral deficiency on the physiological processes that regulate growth (Kozłowski & Pallardy 1997a). Furthermore, potentially toxic compounds, including sulfides, methane, ferrous iron and other reduced compounds that accumulate in poorly aerated soils, may inhibit plant growth (Rosen & Carlson 1984, Kozłowski & Pallardy 1997a, 1997b).

Regeneration of forest stands

Experiments conducted in the field and on the growth of seedlings on cores of severely compacted forest soils show that soil compaction adversely affects stand regeneration by inhibiting seed germination and seedling growth and by inducing seedling mortality. The emergence of seedlings is influenced by mechanical properties of the soil overlying seeds. Soil mechanical impedance greater than 75 kPa inhibited the emergence of *Gossypium hirsutum* seedlings and no seedlings emerged when impedance exceeded 200 kPa (Bowen 1966, 1981).

Growth of tree seedlings on skid trails in forests was reduced by 40% or more over growth in adjacent cutover sites (Pomeroy 1949, Youngberg 1959, Foil & Ralston 1967). A soil BD of 1.8 g cm⁻³ reduced the establishment of *Pinus rigida*, *Pinus nigra* and *Picea abies* seedlings. Penetration of the roots of *P. nigra* seedlings was severely restricted at a BD of 1.4 g cm⁻³ on a silt loam and 1.6 g cm⁻³ on a sandy loam soil (Zisa et al. 1980). After 16 days the root elongation of *Picea abies* and *Pinus sylvestris* seedlings varied inversely with the degree of soil compaction. The growth of *Picea abies* roots was impeded more than that of *Pinus sylvestris* roots (Wästerlund 1985). Root penetration of soil cores by 35–45-day-old *Pseudotsuga menziesii* seedlings decreased linearly with increased soil BD. When downwards growth was impeded, most roots grew laterally in the uncompacted surface soil to a greater length than they grew downwards at the lowest soil BD (Heilman 1981). In compacted soil, dry weight increases shifted to the lateral roots when the growth of taproots of *Ailanthus altissima* was inhibited. In compacted loam the tap roots were twisted, gnarled and often split (Pan & Bassuk 1985). Soil compaction decreased the elongation of the main root, first- and second-order lateral roots and total root length of *Pinus radiata*, but increased the number of first-order lateral roots as well as the diameters of the first-order laterals and the main axis (Sands & Bowen 1978). In Sweden 14–25% of *Picea abies* seedlings showed a 25% reduction in annual height growth because of appreciable soil compaction as well as possible damage to the roots (Wästerlund 1988). Nine measures of growth of *Pinus contorta* and *Picea glauca* seedlings were reduced by soil compaction. These included maximum root depth, maximum root depth in soil cores, total plant weight, shoot weight, root weight, stem diameter, shoot height, seedling survival and shoot weight:root weight ratio (Corns 1988).

Soil compaction restricted the establishment and growth of *Pinus taeda* seedlings during the first 28 weeks after sowing. Seedling heights were lower when soil BD exceeded 1.3 Mg m^{-3} on a sandy clay loam and 1.4 Mg m^{-3} on a loamy sand. The depth of rooting as well as the weights of shoots and roots were reduced on both soils. When BD was less than 1.6 Mg m^{-3} seedling establishment exceeded 90%. However, when BD increased to 1.9 Mg m^{-3} seedling establishment decreased by 44–46% (Tuttle et al. 1988). Root and shoot masses, root system surface areas and root collar diameters of *P. taeda* seedlings grown in pots decreased with increasing soil BD. Significant reductions in height growth were evident within 7 weeks after planting and the inhibition of growth continued until the seedlings were harvested at 19 weeks (Mitchell et al. 1982). Following logging in Australia both the height and diameter growth of 1-year-old *Eucalyptus regnans* seedlings showed a linear, negative correlation with soil BD (Rab 1994).

High rates of seedling mortality on severely compacted soils have been well documented. Approximately 60% of *P. taeda* seedlings died because the radicles could not penetrate the compacted surface soil (Pomeroy 1949). Stocking of *P. taeda* in the lower Coastal Plain decreased from 64000 seedlings ha^{-1} on undisturbed sites to 30000 seedlings ha^{-1} on skid trails (Hatchell et al. 1970). In another study the number per hectare of *P. taeda* trees was reduced by 88–91% by soil compaction (Lockaby & Vidrine 1984). Skid roads produced 74% less volume and had 41% fewer trees than the adjacent forest. Most trees in the skid roads were either suppressed or likely to become suppressed (Wert & Thomas 1981). In Belize soil compaction on roads on which *Swietenia macrophylla* logs were skidded reduced seedling survival (Whitman et al. 1997).

Growth of plants beyond the seedling stage

Soil compaction typically slows plant growth, with inhibition of root growth preceding the effects on shoot and diameter growth (Carmi et al. 1983, Masle & Passioura 1987, Tardieu et al. 1991). Deterioration of tree crowns and reduction of growth in stem diameter may not be apparent until a few years after the soil becomes heavily compacted (Yingling et al. 1979).

Root growth

Roots generally grow through existing pores or by moving soil particles aside. An individual root can enter only rigid soil pores with a diameter greater than that of the root. In many soils, however, roots can squeeze through smaller pores. When roots cannot exert enough pressure to expand small-diameter pores their direction of growth is deflected.

Resistance to root growth typically increases with increasing soil strength (resistance to deformation and/or fracture under applied stress). Soil strength is determined by the sizes and shapes of soil particles, cementing materials, soil organic matter, soil moisture content and soil mineralogy (Young & Bengough 1989). For most plants there is a non-linear inverse relation between the rate of root elongation and soil strength (Misra & Gibbons 1996). As a root enters a soil of high strength, its rate of elongation declines while radial expansion increases behind the root tip. The lowered rate of root elongation reflects a decreased rate of cell division in the meristem together with a decrease in cell elongation. The greater diameters of mechanically impeded roots are associated with increased thickness of the cortex, which results from increases in the diameters of the outer cells as well as increases in the number of cells per unit length of root (Bengough & Mullins 1990).

Severe compaction of soil not only shortens and thickens roots but also may alter their branching patterns. In general, the roots growing in compacted soil become more branched and form more lateral roots. Some species form a mat of proliferated short roots at a shallow soil depth (Gilman et al. 1987). After 76 days the lateral roots of *Ailanthus altissima* seedlings comprised 50 and 24% of the total root dry weight in severely compacted and uncompacted sand, respectively. The corresponding values for compacted and uncompacted loam soils were 70 and 34%, respectively (Pan & Bassuk 1985).

In general, the upper limits of BD at which roots do not penetrate wet soils vary from about 1.4 to 1.6 g cm^{-3} for fine-textured soils to 1.75 g cm^{-3} for coarser-textured soils. However, there is no critical BD that prevents penetration by roots in all soils (Cassel 1983). With an increase in BD and soil strength the total length of primary and lateral roots of *Eucalyptus nitans* seedlings decreased. The average length of lateral roots and diameters of both primary and lateral root-tips increased (Misra & Gibbons 1996). The capacity for root growth in compacted soil often varies among plant species. For example, root

growth of six species of trees varied appreciably in compacted soil. *Larix sibirica* and *Quercus robur* were the most capable of forming roots in compacted soil (up to a BD of 1.89 and 1.84 g cm⁻³, respectively). *Picea abies* and *Tilia cordata* were the least capable of initiating roots (to a BD of 1.61 and 1.55 g cm⁻³, respectively). Intermediate BD values were obtained for root formation of *Pinus sylvestris* and *Betula pendula* (Korotaev 1992).

Root growth is greatly influenced by the distribution of root channels (Wang et al. 1986, Van Noordwijk et al. 1991), structural cracks and macropores (> 1000 µm) as well as the largest mesopores (10–1000 µm) in compacted soils. These pores consist largely of holes in aggregate interped pores and root or worm channels (Wang et al. 1986, Edwards & Bohlen 1996). Roots often grow into such cracks and pores, thereby tapping a larger reservoir of water and mineral nutrients. In very compacted soils root growth may be confined almost entirely to cracks and cleavage planes (Taylor et al. 1966). When a crop of high root penetrating capacity is followed by a crop of low penetrating capacity, root channels created by the earlier crop may result in deeper rooting of the subsequent crop. For example, where porosity of compacted layers in the subsoil was increased by *Paspalum notatum* and *Festuca arundinacea*, the *Gossypium hirsutum* plants that followed responded to the increased porosity by deeper root penetration and greater yield (Hullugalle & Lal 1986).

Growth of roots in soil pores varies appreciably with pore size. A few large biopores were less favorable for root growth than a network of small diameter pores made by the roots of other plants. Some radicles died in large, empty pores. The more vigorous root growth in small pores was attributed to better soil contact and less compression of the soil by growing roots (Stirzaker et al. 1996).

Not all parts of a root system are equally exposed to soil compaction. Hence, because of the compensatory growth of unimpeded roots, the distribution of root growth and not the total length of roots sometimes is altered (Unger & Kaspar 1994). Paths of low mechanical impedance are often created by soil fauna, particularly earthworms, and smaller pores by roots of preceding crops (Ehlers et al. 1983, Bennie 1991). Subsoil compaction inhibited the penetration of soil by *Pinus radiata* roots and increased water stress in trees. However, the effects of compaction on root growth were essentially negated when the subsoil was perforated to simulate old-root channels and

convert 0.2% of the soil volume into vertical channels (Nambiar & Sands 1992).

The capacity of roots to penetrate soil surfaces decreased with increasing soil strength and increasing angle of the surface relative to the horizontal. However, the responses of individual roots varied considerably. Roots elongated more rapidly in cracks than through undisturbed clods, provided the cracks were not oriented at an oblique angle to the normal geotropic direction of root growth (Whiteley & Dexter 1983). The proportion of roots that penetrated the interface between soil layers varied with the strength of these layers, the maximum pressure that could be exerted by roots to displace soil particles and the angle at which the roots approached the compacted layer (Materechera et al. 1992).

Mechanical impedance and the O₂ content of the soil often interact in their influence on root growth (Gill & Mittler 1956, Day & Bassuk 1994). At moderate BD of soil (1.5–1.6 g cm⁻³) penetration of soil by roots was inhibited when the O₂ content decreased to less than 10%. At higher levels of BD mechanical impedance appeared to be the dominating factor affecting root growth (Tackett & Pearson 1964). At low levels of soil compaction, root penetration of soil strongly depended on O₂ content, increasing from negligible penetration at 3% O₂ to maximal root growth at 21% O₂. In general, soil compaction had a strong inhibitory effect on root penetration when the O₂ level was high and no significant effect at a low O₂ level (Hopkins & Patrick 1969).

Where runoff of water does not occur from compacted soil and surface water accumulates many roots may die and decay because of increased activity of *Phytophthora* fungi, which tolerate low soil aeration (Duniway & Gordon 1986, Gray & Pope 1986). These organisms are stimulated by the attraction of their zoospores to root exudates, including sugars, amino acids and ethanol (Stolzy & Sojka 1984). A lack of fibrous roots in fine-textured, wet soils was associated with root decay by fungi as well as the inhibition of root growth because of O₂ deficiency to the roots (Stolzy et al. 1965). The importance of O₂ deficiency to the development of root rot is emphasized by many studies showing the extensive death and decay of roots in flooded soils (for reviews see Kozlowski 1982b, 1984a, 1984b, 1985, 1986, 1997). Root growth may also be inhibited by a variety of potentially toxic compounds produced by roots and anaerobic bacteria in flooded soils (Ponnamperuma 1984).

There are many examples of the inhibitory effects of soil compaction on the growth of trees beyond the seedling stage and only a few examples will be given here. Increasing compaction of a sandy nursery soil from an optimum (1.2 g cm^{-3}) to $1.5\text{--}1.6 \text{ g cm}^{-3}$ by machinery led to losses in dry matter production of nearly 50% (Thithamer 1989). In *Pinus ponderosa* forests soil densities on skid trails at 8- and 15-cm depths and 23- or 30-cm depths were 18 and 9% greater, respectively, than those of undisturbed soils. Sixteen years later the growth of moderately impacted trees was reduced by 6% and the growth of heavily impacted trees by 12% (Froehlich 1979). In another study, the height, diameter and volume growths of *P. ponderosa* were reduced by 5, 8 and 20%, respectively (Froehlich et al. 1986). Height growth in a 16-year-old *P. ponderosa* plantation was 43% less and 13% less at age 15 years in areas of highest BD than where BD was lowest (Helms et al. 1986). The volume increment of 16-year-old *P. ponderosa* trees per unit of land area was significantly reduced on skid trails (Helms & Hipkin 1986). Modeling predicted that the compaction of soil by typical harvesting practices would reduce the total volume yield of *Pseudotsuga menziesii* stands by 8.1% in the second rotation (Stewart et al. 1988).

In New Zealand the height growth of *Pinus radiata* trees on skid trails was 32% less than off the skid trails. Survival and tree form were also poorer on the skid trails (Murphy 1983). The reduction of growth and change in form of *P. radiata* on skid trails influenced the selection of trees for thinning and pruning. Hence, when the trees were 7.5 years old only 4% of those planted on skid trails formed the final crop, compared with 15% of those planted off the skid trails (Firth & Murphy 1989). In Australia tree survival was lower and the basal area reduced by approximately half on severely compacted soils. *Gmelina arborea* grew better than *Terminalia brassica* on compacted soil (Cheatle 1991). In Costa Rica the diameter growth of trees on abandoned logging roads was estimated to take at least 80 years to reach the level found in logged forests and the recovery of species richness took even longer (Guariguata & Dupuy 1997).

Not all investigators were able to show a growth reduction of trees as a result of soil compaction in areas of timber harvesting. For example, Miller et al. (1996) noted that on skid trails, initial increases in BD of 40% or more did not affect the 8-year growth or survival of *Pseudotsuga menziesii* and *Picea sitchensis* trees. Initially favorable soil conditions and subse-

quently favorable climatic conditions compensated for the potentially adverse effects of compaction of silt loam soils.

Reproductive growth

Appreciable soil compaction typically decreases both the quality and yield of fruit crops. Ground cover vegetation improved the production of *Prunus persica* fruits compared with soil cultivation. Ground cover vegetation appeared to increase the uptake of mineral nutrients. The improved yield of fruits appeared to be due mainly to the effects on soil resistance to root penetration (Bussi et al. 1994). Interrow soil compaction by tractor traffic resulted in a lower yield of *Rubus idaeus* fruits and lower specific fruit weights (Wieniarska et al. 1987). The reduction in yield by soil compaction was greater for fruits of *Rubus idaeus* than for those of *Ribes schlehtendaltii* or *R. nigrum* (Kesik & Stanek 1983).

In the Mississippi River Delta the yields of *Gossypium hirsutum* bolls and lint were often appreciably reduced by surface compaction and subsoil pans (Lowry et al. 1970). Yields of *G. hirsutum* bolls were increased by controlling vehicle traffic (Dumas et al. 1973, Hadas et al. 1985). Stunted *G. hirsutum* plants affected by tillage pans had few viable fruiting sites and produced few harvestable bolls. Shallow tillage pans were more detrimental than deep tillage pans to *Gossypium* development (McConnell et al. 1989).

AMELIORATION OF SOIL COMPACTION

The deleterious effects of soil compaction are not completely eliminated by natural processes such as freezing and thawing, wetting and drying, non-uniform water absorption and soil dehydration by root-system uptake. A number of treatments has been used with variable success in alleviating the harmful effects of soil compaction on plant growth (Pittenger & Stamen 1990). These include planting of species that can tolerate soil compaction (Bowen 1981, Ruark et al. 1982), controlling vehicle traffic (Gerik et al. 1987, Marsh et al. 1996), amending or replacing soil (Jim 1993), loosening soil (Spoor 1980, Voorhees 1983, Day & Bassuk 1993, Murphy et al. 1993, Unger & Kaspar 1994), installing drainage systems (Harris 1992), trenching (Day et al. 1995) and applying fertilizers (Sheriff & Nambiar 1995). Prevention of soil compaction before planting is generally much preferred over post-planting practices because the latter are expensive, difficult to apply, sometimes ineffective and

may injure roots (Howard et al. 1981). Improving the physical properties of compacted soils is often difficult because only a part of the soil deformation can be reversed (Horn et al. 1995).

Species that adjust readily to soil fill include *Ailanthus altissima*, *Gleditsia triacanthos*, *Pinus taeda*, *Robinia pseudoacacia*, *Ulmus americana*, *U. parvifolia* and a number of flood-tolerant species (Kozlowski 1984a, 1984b, 1984c, 1986). Species considered intolerant of poor soil aeration include *Cornus* spp., *Liriodendron tulipifera*, *Fagus sylvatica*, *Picea* spp. and *Pinus resinosa* (Ruark et al. 1982).

A variety of organic materials has been used in soil mixes and backfills, including peats, composts, organic wastes and mulches (Bowen 1981, Corley 1984, Donnelly & Shane 1986, Soane 1990, Jim 1993). Increases in organic matter reduce soil compactibility and more so at high than at low soil moisture contents. Roots and fungal hyphae provide a filamentous network that resists compactive loads. Organic matter reduces soil compactibility by increasing resistance to deformation and increasing soil elasticity. Even small changes in the amount of organic matter may appreciably influence soil compaction (Wingate-Hill & Jakobson 1982, Soane 1990). The activity of microbial populations induced by organic matter may antagonize root diseases (Avnimelech 1986).

Organic materials have been used routinely in the sand-based root zones of sports turf soils to increase retention of water and mineral nutrients and decrease soil BD (Baker 1984, Baker & Hacker 1988, McCoy 1992). Application of organic wastes increased the C content of soil which, in turn, decreased BD and increased both the water-holding capacity and hydraulic conductivity. In addition, organic matter and soil aggregation were inversely related to runoff volumes and sediment losses. Hence, increased aggregation led to less soil erosion (Khaleel et al. 1981). The effects of peat on the compressibility of soil vary with the type of peat. Fibrous peat, which holds a great deal of water in a network of macropores, is more compressible than amorphous peat (Soane 1990).

Soil mixes have been used advantageously with a matrix of coarse materials to provide a skeleton that resists soil deformation. In heavily used recreation areas, such as playgrounds and picnic sites, soil compaction has been counteracted by rotary tilling of lightweight aggregates into the soil. Kopinga (1985a) used a coarse lava slag in a 2:1 lava:soil ratio. Patterson (1975, 1976) evaluated four soil amendments: (1) sintered fly ash; (2) expanded slate; (3) coarse con-

struction sand; and (4) digested sewage sludge. A mixture with 33% by volume of expanded slate had the lowest BD and most pore space. Sintered fly ash had less effect on soil aeration than the expanded slate. Coarse sand had poorer physical attributes than either of the two aggregates.

Surface compaction has often been alleviated by spike and core aeration and by rototilling; subsurface compaction by deep jetting, subsoiling and trenching (Spoor 1980, Mason & Cullen 1986, Rolf 1991, Jim 1993, Unger & Kaspar 1994). Both hollow-tine cultivation (HTC) and solid-tine cultivation (STC) ameliorated the adverse effects of soil compaction. Saturated water conductivity and air porosity were 49 and 21% greater, respectively, with HTC than with STC. Under wet soil conditions HTC yielded the highest turf quality. Although STC provided some short-term benefits it required repeated applications and tended to develop a cultivation pan (Murphy et al. 1993).

Subsoiling or deep-tillage has been used to decrease soil BD. However, subsoiling implements may injure roots and often have deleterious long-term effects on root growth (Kliejunas 1992, Otrosina et al. 1996). In Idaho subsoiling increased water infiltration by 10% and decreased erosion by 278% (Sojka et al. 1993). In Malaysia plowing or subsurface ripping of log landings and skid trails before planting of trees reduced soil compaction. This was done by attaching tines or rippers to bulldozers that were used for skidding of logs (Nussbaum & Hoe 1990). Subsoiling treatments typically have been used when the soil is dry enough to the depth to which it should be treated. Subsoiling treatments have not been very effective when applied to sites that did not require them, when subsoiling did not adequately loosen the soil or when soil loosening was not stable (Rolf 1994).

The BD of a sandy loam was decreased by pneumatic subsoil loosening with a soil aerator, and porosity, saturated hydraulic conductivity and air permeability were increased. The ameliorating effects were less obvious on a loam soil (Rolf 1992). The effects of soil loosening by jets of compressed air on height growth of trees have been variable, depending to a considerable extent on soil type and soil moisture content. Compressed air injection into soil increased height growth of *Betula* sp. by 21–37% in different years (Hodge 1993). Whereas shoot growth of *Platanus x hispanica* was stimulated 2 years after compressed air was injected into the soil, there was no clear effect on *Aesculus hippocastanum* growing on a

clay loam (Hodge 1991). For deeply compacted soil a combination of deep loosening of soil and drainage was more efficient than either treatment alone (Weyers et al. 1993).

The installation of drainage systems serves several purposes, including the prevention of soil anaerobiosis, accelerating drying of soil to a tolerable load-bearing capacity and protecting soils against runoff (Castle et al. 1984, Adams 1986). Subsoiling also may improve drainage (Trowse 1983). Subsoil-perforated pipes connected to vertical shafts opening to the atmosphere have been installed to facilitate soil aeration (Kopinga 1985b, Harris 1992, Jim 1993, Larson et al. 1994). Problems and methods of installing drainage systems are discussed by the Ministry of Agriculture, Fisheries and Food (Anon. 1983), Castle et al. (1984), Adams (1986) and Harris (1992).

Trenches that radiate outwards from the bases of tree stems and are located to avoid major roots have been useful in alleviating the harmful effects of severe soil compaction on tree growth. Such trenches typically have been filled with soil mixed with organic matter. Backfilled trenches in a Harlington Clay soil increased water infiltration, decreased soil BD in the trenches and increased the volume of soil available to roots. Penetration of roots was increased from 60 to 122 cm in 102-cm-deep trenches (Heilman & Gonzalez 1973). Day et al. (1995) studied the effects of four treatments: (1) peat-amended backfills; (2) vertical drainage mat panels; (3) radiating trenches filled with sandy loam; and (4) gravel-filled sump drains. Shoot growth of *Pyrus calleryana* was stimulated most by treatments that reduced the mechanical impedance of the soil (trenches and amended backfills) and least by treatments that did not (vertical drains). Soil trenches and amended backfills accounted for 82 and 73% more shoot growth, respectively, during the second growing season.

The harmful effects of subsoil compaction on tree growth sometimes have also been alleviated by judicious fertilizer applications. Fertilized trees growing in a compacted subsoil had more root biomass in the upper 0.3 m of the soil profile, more available water and higher growth rates than unfertilized trees (Sheriff & Nambiar 1995).

ACKNOWLEDGMENTS

The valuable assistance of Joy and Pavel Svihra in manuscript preparation is gratefully acknowledged.

REFERENCES

- Adams, W. A. 1986. Practical aspects of sports field drainage. *Soil Use Manage.* 2: 51–54.
- Ahlgren, C. F. & Ahlgren, I. F. 1960. Ecological effects of forest fires. *Bot. Rev.* 17: 483–533.
- Alberty, C. A., Pellett, H. M. & Taylor, D. H. 1984. Characterization of soil compaction at construction sites and woody plant responses. *J. Environ. Hort.* 2: 48–53.
- Allamaras, R. R., Kraft, J. M. & Miller, D. E. 1988. Effects of soil compaction and incorporated crop residue on root health. *Annu. Rev. Phytopathol.* 26: 219–244.
- Anderson, J. M. & Spencer, T. 1991. Carbon, nutrient and water balances of tropical rain forest ecosystems subject to disturbance: management implications and research proposals. *Man Biosphere Digest* 7, UNESCO, Paris.
- Anon. 1983. Ministry of Agriculture, Fisheries and Food. Technical Notes on Workmanship and Materials for Drainage Schemes BL 4316. HMSO, London.
- Arah, J. R. M. & Smith, K. A. 1989. Steady-state denitrification in aggregated soils: a mathematical model. *J. Soil Sci.* 40: 139–149.
- Arend, J. L. 1941. Infiltration rates of forest soils in the Missouri Ozarks as affected by woods burning and litter removal. *J. For.* 39: 726–728.
- Arvidsson, J. & Jokela, W. E. 1995. A lysimeter study of soil compaction effects on evapotranspiration in a barley crop. *Swed. J. Agric. Res.* 25: 109–118.
- Assouline, S., Tavares-Filho, J. & Tessier, D. 1997. Effect of soil compaction on soil physical and hydraulic properties: experimental results and modeling. *Soil Sci. Soc. Am. J.* 61: 390–398.
- Avnimelech, Y. 1986. Organic residues in modern agriculture. In Chen, Y. & Avnimelech, Y. (eds.). *The Role of Organic Matter in Modern Agriculture*, pp. 1–10. Martinus Nijhoff, Dordrecht.
- Baker, S. W. 1984. Long-term effects of three amendment materials on the moisture retention characteristics of a sand–soil mix. *J. Sports Turf Res. Inst.* 60: 61–65.
- Baker, S. W. & Hacker, J. W. 1988. The use of peat in a Prunty-Mulqueen sand carpet construction: effects of application rate and depth. *J. Sports Turf Res. Inst.* 64: 87–98.
- Ball, B. C. & Smith, K. A. 1991. Gas movement. In Smith, K. A. & Mullins, C. E. (eds.). *Soil Analysis: Physical Methods*, pp. 511–549. Marcel Dekker, New York.
- Barber, S. A. 1995. *Soil Nutrient Bioavailability*. Wiley Interscience, New York.
- Bengough, A. G. & Mullins, C. E. 1990. Mechanical impedance to root growth: a review of experimental techniques and root growth responses. *J. Soil Sci.* 41: 341–358.
- Bennie, A. T. P. 1991. Growth and mechanical impedance. In Waisel, Y., Eshel, A. & Kafkafi, U. (eds.). *Plant Roots: The Hidden Half*, pp. 393–414. Marcel Dekker, New York.
- Bezkorowajnyi, P. G., Gordon, A. M. & McBride, R. A. 1993. The effect of cattle foot traffic on soil compaction in a silvopastoral system. *Agrofor. Syst.* 21: 1–10.

- Boll, B. C. 1987. Air permeability and gas diffusion. In Monnier, G. & Goss, M. J. (eds.). *Soil Compaction and Vegetation*, pp. 15–23. Balkema, Rotterdam.
- Bowen, G. D. 1984. Tree roots and the use of soil nutrients. In Bowen, G. D. & Nambiar, E. K. S. (eds.). *Nutrition of Plantation Forests*, pp. 147–179. Academic Press, London.
- Bowen, H. D. 1966. Measurement of edaphic factors for determining planter specifications. *Trans. Am. Assoc. Agricult. Eng.* 9: 725–735.
- Bowen, H. D. 1981. Alleviating mechanical impedance. In Arkin, A. F. & Taylor, H. M. (eds.). *Modifying the Root Environment to Reduce Stress*, Monograph 4, pp. 21–57. American Society of Agricultural Engineers, St Joseph, MO.
- Boyer, J. M. & South, D. B. 1988. Loblolly pine seedling morphology and production at 53 southern forest nurseries. US Department of Agriculture, Forest Service Tree Planters Note 13.
- Broder, M. W., Moran, J. W., Peterson, G. A. & Fenster, C. R. 1984. Fallow tillage influence on spring populations of soil nitrifiers, denitrifiers, and available nitrogen. *Soil Sci. Soc. Am. J.* 48: 1060–1067.
- Burgess, T. I., Malajczuk, N. & Grove, T. S. 1993. The ability of 16 ectomycorrhizal fungi to increase growth and phosphorus uptake of *Eucalyptus globulus* Labill. and *E. diversicolor* F. Muell. *Plant Soil* 153: 155–164.
- Bussi, G., Huguet, J. G., Besset, J. & Girard, T. 1994. The consequences of cultural practices on growth and cropping of peach trees subjected to localized irrigation: effects of soil management, nitrogen fertilization and tree density. *J. Hort. Sci.* 69: 905–914.
- Cannell, R. Q. 1977. Soil aeration and compaction in relation to root growth and soil management. *Appl. Biol.* 2: 1–86.
- Carmi, A., Hesketh, J. D., Enos, W. T. & Peters, D. B. 1983. Interrelationships between shoot growth and photosynthesis as affected by root growth restriction. *Photosynthetica* 17: 240–245.
- Carrow, R. N. 1981. Soil compaction, *Golf Course Manage.* 49: 59–62, 66.
- Carrow, R. N. & Petrovic, A. M. 1992. Effects of traffic on turf grasses. In Carrow, R. N. & Shearman, R. C. (eds.). *Turfgrass*, pp. 285–330. American Society of Agronomy, Madison, WI.
- Carter, M. R. 1990. Relative measures of soil bulk density to characterize compaction in tillage studies on fine sandy loams. *Can. J. Soil Sci.* 70: 425–434.
- Cassel, D. K. 1983. Effects of soil characteristics and tillage practices on water storage and its availability to plant roots. In Raper, C. D., Jr. & Kramer, P. J. (editors). *Crop Relations to Water and Temperature Stresses in Humid Temperate Climates*, pp. 167–186. Westview Press, Boulder, CO.
- Castilo, S. R., Dowdy, R. H., Bradford, J. M. & Larson, W. E. 1982. Effects of applied mechanical stress on plant growth and nutrient uptake. *Agron. J.* 74: 526–530.
- Castle, D. A., McCunnall, J. & Ting, I. M. 1984. *Field Drainage Principles and Practices*. Academic Press, London.
- Cheatle, R. J. 1991. Tree growth on compacted Oxisol. *Soil Tillage Res.* 19: 331–344.
- Chen, Y., Tarchitzky, J., Brouvier, J., Morin, J. & Bonin, A. 1980. Scanning electron microscope observations on soil crusts and their formation. *Soil Sci.* 130: 49–55.
- Clements, R. O. 1982. Some consequences of large and frequent pesticide applications to grassland. In Lee, K. (ed.). *Proceedings of the Third Australian Conference on Grassland Ecology*, pp. 393–396. South Australian Government Printer, Adelaide.
- Cochran, P. H. & Brock, T. 1985. Soil compaction and initial height growth of planted ponderosa pine. USDA Forest Service, Pacific Northwest Forest and Range Experimental Station Research Note PSW 434, Portland, OR.
- Cole, D. N. 1982. Wilderness campsite impacts: effect of amount of use. USDA Forest Service Res. Pap. INT-284. Ogden, Utah.
- Cole, D. N. 1983. Campsite conditions in the Bob Marshall Wilderness, Montana. USDA Forest Service Intermountain Experiment Station Research Paper INT-312, 1–18.
- Cole, D. N. 1986. Recreational impacts on backcountry campsites in Grand Canyon National Park, Arizona, USA. *Environ. Manage.* 10: 651–659.
- Cole, D. N. & Fichtler, R. K. 1983. Campsite impact on three western wilderness areas. *Environ. Manage.* 7: 275–288.
- Comerford, N. B., Kidder, G. K. & Mollitor, A. V. 1984. Importance of subsoil fertility to forest and non-forest plant nutrition. In Stone, E. L. (ed.). *Proceedings, Sixth North American Forest Soils Conference*, pp. 381–404. University of Tennessee, Knoxville, TN.
- Conlin, T. S. S. & van den Driessche, R. 1996. Short-term effects of soil compaction on growth of *Pinus contorta* seedlings. *Can. J. For. Res.* 26: 727–739.
- Corley, W. L. 1984. Soil amendments at planting. *J. Environ. Hort.* 2: 27–30.
- Corns, G. W. 1988. Compaction by forestry equipment and effects on coniferous seedling growth on four soils in the Alberta foothills. *Can. J. For. Res.* 18: 75–84.
- Crane, J. H. & Davies, F. S. 1989. Flooding response of *Vaccinium* species. *Hort. Sci.* 24: 203–210.
- Cullen, S. J., Montagne, C. & Ferguson, H. 1991. Timber harvest trafficking and soil compaction in western Montana. *Soil Sci. Soc. Am. J.* 55: 1416–1421.
- Currie, J. A. 1984. Gas diffusion through soil crumbs: the effects of compaction and wetting. *J. Soil Sci.* 35: 1–10.
- Danell, E. 1994. Formation and growth of the ectomycorrhiza of *Cantharellus cibarius*. *Mycorrhiza* 5: 89–97.
- Davis, R. R. 1952. Physical condition of putting green soils and other environmental factors affecting greens. *USGA J. Turf Manage.* 6: 25–27.
- Davis, S. 1992. Bulk density changes in two central Oregon soils following tractor logging and slash piling. *West. J. Appl. For.* 7: 86–88.
- Day, S. D. & Bassuk, N. L. 1993. Evaluation of remedial drainage aeration and soil-loosening techniques on two landscape trees in compacted soil. *HortScience* 28: 127.

- Day, S. D. & Bassuk, N. L. 1994. A review of the effects of soil compaction treatments on landscape trees. *J. Arboricult.* 20: 9–17.
- Day, S. D., Bassuk, N. L. & van Es, H. 1995. Effects of four compaction remediation methods for landscape trees on soil aeration, mechanical impedance and tree establishment. *J. Environ. Hort.* 13: 64–71.
- De Bano, L. F. 1981. Water repellent soils: a state of the art. USDA Forest Service General Technical Report PSW-PSW-46.
- Demko, J. 1994. Compacting of forest soils during wood skidding (special forest wheel tractor LKT 40). *Lesnictvi (Prague)* 40: 482–485.
- Deschênes, C. 1989. Pression exercée sur le sol par la machinerie forestière: repertoire Ministère de l'Énergie et des Ressources. Canada, Quebec (In French).
- Dickerson, B. P. 1976. Soil compaction after tree length skidding in northern Mississippi. *Soil Sci. Soc. Am. J.* 40: 965–966.
- Dolan, M. S., Dowdy, R. H., Voorhees, W. B., Johnson, J. F. & Bidwell-Schrader, A. M. 1992. Corn phosphorus and potassium uptake in response to soil compaction. *Agron. J.* 84: 639–642.
- Donnelly, J. R. & Shane, J. B. 1986. Forest ecosystem response to artificially induced soil compaction. I. Soil physical properties and tree diameter growth. *Can. J. For. Res.* 16: 750–754.
- Dotzenko, A. K., Papamichos, N. T. & Romine, D. S. 1967. Effect of recreational use on soil and moisture conditions in Rocky Mountain National Park. *J. Soil Water Conserv.* 22: 196–197.
- Douglas, J. T. & Crawford, C. E. 1993. The response of a ryegrass sward to wheel traffic and applied nitrogen. *Grass Forage Sci.* 49: 99–100.
- Dumas, W. T., Trowse, A. C., Smith, L. A., Kummar, F. A. & Gill, W. R. 1973. Development and evaluation of tillage and other cultural practices in a controlled traffic system for cotton in the southern coastal plains. *Trans. Am. Soc. Agric. Eng.* 16: 872–875.
- Duniway, J. M. & Gordon, T. R. 1986. Water relations and pathogen activity in soil. In Ayres, P. L. & Boddy, L. (eds.). *Water, Fungi and Plants*, pp. 119–137. Cambridge University Press, Cambridge.
- Edwards, C. A. & Bohlen, P. J. 1992. The effects of toxic chemical on earthworms. *Rev. Environ. Contam. Toxicol.* 125: 23–99.
- Edwards, C. A. & Bohlen, P. J. 1996. *Biology and Ecology of Earthworms*, 3rd edn. London: Chapman and Hall.
- Edwards, L. M. & Burney, J. R. 1989. The effect of antecedent freeze–thaw frequency on runoff and soil loss from frozen soil with and without soil compaction and ground cover. *Can. J. Soil Sci.* 69: 799–812.
- Edwards, L. M. & Burney, J. R. 1991. Sediment concentration of interrill runoff under varying soil, ground cover, soil compaction, and freezing regimes. *J. Environ. Qual.* 20: 403–407.
- Ehlers, W., Köpke, U., Hesse, E. & Böhm, W. 1983. Penetration resistance and root growth of oats in tilled and untilled loess soil. *Soil Tillage Res.* 3: 261–275.
- Else, M. A., Tiekstra, A. E., Croker, S. J., Davies, W. J. & Jackson, M. B. 1996. Stomatal closure in flooded tomato plants involves abscisic acid and a chemically unidentified anti-transpirant in xylem sap. *Plant Physiol.* 112: 239–247.
- Entry, J. A., Reeves, D. W., Mudd, E., Lee, W. J., Guertel, E. & Raper, R. L. 1996. Influence of compaction from wheel traffic and tillage on arbuscular mycorrhizal infection and nutrient uptake by *Zea mays*. *Plant Soil* 180: 139–146.
- Epron, D., Godard, D., Cornic, G. & Gentry, B. 1995. Limitation of net CO₂ assimilation rate by internal resistances to CO₂ transfer in the leaves of two tree species (*Fagus sylvatica* L. and *Castanea sativa* Mill.). *Plant Cell Environ.* 18: 43–51.
- Federer, C. A., Tenpas, G. H., Schmidt, D. R. & Tanner, C. B. 1961. Pasture compaction by animal traffic. *Agron. J.* 53: 53–54.
- Fedotov, A. I. 1989. Air permeability of the arable layer of peat soil with different compaction. *Pochvovedenia* 9: 122–126.
- Ferrero, A. F. 1991. Effect of compaction simulating cattle trampling on soil physical characteristics in woodland. *Soil Tillage Res.* 19: 319–329.
- Filer, T. H. 1975. Mycorrhizal and soil microflora in a green tree reservoir. *For. Sci.* 24: 36–39.
- Firth, J. & Murphy, G. 1989. Skidtrails and their effects on the growth and management of young *Pinus radiata*. *N Z J. For. Sci.* 19: 22–28.
- Fisher, H. M. & Stone, E. L. 1990a. Air-conducting porosity in slash pine roots from saturated soils. *For. Sci.* 36: 18–33.
- Fisher, H. M. & Stone, E. L. 1990b. Active potassium uptake by slash pine roots from O₂-depleted solutions. *For. Sci.* 36: 582–598.
- Foil, R. R. & Ralston, C. W. 1967. The establishment and growth of loblolly pine seedlings on compacted soils. *Soil Sci. Soc. Am. Proc.* 31: 565–568.
- Frede, H. G. 1987. The importance of pore volume and pore geometry to soil aeration. In Monnier, G. & Gross, M. J. (eds.). *Soil Compaction and Regeneration*, pp. 25–29. Balkema, Rotterdam.
- Froehlich, H. A. 1979. Soil compaction from logging equipment. Effects on growth of young ponderosa pine. *J. Soil Water Conserv.* 34: 276–278.
- Froehlich, H. A. & McNabb, D. H. 1984. Minimizing soil compaction in Pacific Northwest forests. In Stone, E. L. (ed.). *Forest Soils and Treatment Impacts*, pp. 159–192. University of Tennessee, Knoxville, TN.
- Froehlich, H. A., Miles, D. W. R. & Robbins, R. W. 1986. Growth of young *Pinus ponderosa* and *Pinus contorta* on compacted soil in central Washington. *For. Ecol. Manage.* 15: 285–291.
- Gameda, S., Raghaven, G. S. V., McKyes, E., Watson, A. K. & Muhays, G. 1994. Long-term effects of a single incidence of high axle load compaction on a clay soil in Quebec. *Soil Tillage Res.* 29: 179–181.
- Gayoso, J. & Iroume, A. 1991. Compaction and soil disturbance from logging in Southern Chile. *Ann. Sci. For.* 48: 63–71.

- Gerik, T. J., Morrison, J. E., Jr., & Chichester, F. W. 1987. Effects of controlled traffic on soil physical properties and crop rooting. *Agron. J.* 79: 434–438.
- Gifford, G. F. & Hawkins, R. H. 1978. Hydrologic impact of grazing on infiltration: a critical review. *Water Resources Res.* 14: 305–313.
- Gill, W. R. 1971. Economic assessment of soil compaction. In Barnes, K. K., Carleton, W. M., Taylor, H. M., Throckmorton, R. I. & Van den Berg, G. F. (eds.). *Compaction of Agricultural Soils*, pp. 431–458. American Society of Agricultural Engineers, St Joseph, MO.
- Gill, W. R. & Mittler, R. D. 1956. A method for study of the influence of mechanical impedance and aeration on the growth of seedling roots. *Proc. Soil Sci. Soc. Am.* 20: 154–157.
- Gilman, E. F., Leone, I. A. & Flower, F. B. 1987. Effect of soil compaction and oxygen content on vertical and horizontal root distribution. *J. Environ. Hort.* 5: 33–36.
- Goss, M. J. & Russell, R. S. 1980. Effects of mechanical impedance on root growth in barley (*Hordeum vulgare* L.). III. Observations on the mechanism of response. *J. Exp. Bot.* 31: 577–588.
- Grable, A. R. 1971. Effects of compaction on content and transmission of air in soils. In Barnes, K. K., Carleton, W. M., Taylor, H. M., Throckmorton, R. I. & Van den Berg, G. E. (eds.). *Compaction of Agricultural Soils*, pp. 154–164. American Society of Agricultural Engineers, St Joseph, MO.
- Gray, L. E. & Pope, R. A. 1986. Influence of soil compaction on soybean stand, yield and *Phytophthora* root rot incidence. *Agron. J.* 78: 189–191.
- Greacen, E. L. & Sands, R. 1980. Compaction of forest soils: a review. *Aust. J. Soil Res.* 18: 163–189.
- Groffman, P. M. & Tiedje, J. M. 1989. Denitrification in north temperate forest soils: relationships between denitrification and environmental factors at the landscape scale. *Soil Biol. Biochem.* 21: 621–626.
- Groffman, P. M., Tiedje, J. M., Mokma, D. L. & Simkins, S. 1992. Regional scale analysis of denitrification in north temperate forest soils. *Landscape Ecol.* 7: 45–53.
- Guariguata, M. R. & Dupuy, J. M. 1997. Forest regeneration in abandoned logging roads in lowland Costa Rica. *Biotropica* 29: 15–28.
- Hadas, A. 1994. Soil compaction caused by high axle loads: review of concepts and experimental data. *Soil Tillage Res.* 29: 253–276.
- Hadas, A., Wolf, D. & Rawitz, E. 1985. Residual compaction effects on cotton stand and yield. *Trans. Am. Soc. Agric. Eng.* 28: 691–696.
- Haig, I. T. 1938. Fire in modern forest management. *J. For.* 36: 1045–1051.
- Hajabbasi, M. A., Jalalin, A. & Karimzadeh, H. C. 1997. Deforestation effects on soil physical and chemical properties, Lordegan, Iran. *Plant Soil* 90: 301–308.
- Håkansson, I. W. B., Voorhees, P., Elonen, P., Raghavan, G. V. S., Lowery, B., van Wijk, A. L. M., Rasmussen, K. & Riley, H. 1987. Effect of high axle-load traffic on subsoil compaction and crop yield in humid regions with annual freezing. *Soil Tillage Res.* 10: 259–268.
- Hansen, S. & Bakken, L. R. 1993. Nitrous oxide, carbon dioxide and oxygen concentrations in soil air influenced by organic and inorganic fertilizers and soil compaction. *Norweg. J. Agric. Sci.* 7: 1–10.
- Harris, R. W. 1992. *Arboriculture*. Prentice-Hall, Englewood Cliffs, NJ.
- Hatchell, G. E., Ralston, C. W. & Foil, R. R. 1970. Soil disturbances in logging. *J. For.* 68: 772–775.
- He, C. J., Finlayson, S. A., Drew, M. C., Jordan, W. R. & Morgan, P. W. 1996. Ethylene biosynthesis during aerenchyma formation in roots of maize subjected to mechanical impedance and hypoxia. *Plant Physiol.* 112: 1679–1685.
- Hegarty, T. W. & Royle, S. M. 1978. Combined effects of moisture content prior to compaction, comparative effort and rainfall quantity on soil crust strength. *J. Soil Sci.* 29: 167–173.
- Heilman, M. D. & Gonzalez, C. L. 1973. Effect of narrow trenching in Harlingen clay soil on plant growth, rooting depth, and salinity. *Agron. J.* 65: 816–819.
- Heilman, P. 1981. Root penetration of Douglas-fir seedlings into compacted soil. *For. Sci.* 27: 660–666.
- Helms, J. A. & Hipkin, C. 1986. Effects of soil compaction on tree volume in a California ponderosa pine plantation. *West. J. Appl. For.* 1: 121–124.
- Helms, J. A., Hipkin, C. & Alexander, E. B. 1986. Effects of soil compaction on height growth of a California ponderosa pine plantation. *West. J. Appl. For.* 1: 104–108.
- Hillel, D. L. 1982. *Introduction to Soil Physics*. Academic Press, New York.
- Hodge, S. J. 1991. Improving the growth of established amenity trees: site physical conditions. *Arboriculture Research Note 102/91/ARB*. Arboriculture Advisory and Information Service, Farnham.
- Hodge, S. J. 1993. The effect of stem nutrient injection and compressed air soil injection on the performance of established amenity trees. *Arboricult. J.* 17: 287–300.
- Hopkins, R. M. & Patrick, W. H., Jr., 1969. Combined effect of oxygen content and soil compaction on root penetration. *Soil Sci.* 108: 408–413.
- Horn, R., Domzal, H., Slowinska-Jurkiewicz, A. & Van Ouwerkerk, C. 1995. Soil compaction processes and their effects on the structure of arable soils and the environment. *Soil Tillage Res.* 35: 23–36.
- Howard, R. F., Singer, M. J. & Frantz, G. A. 1981. Effects of soil properties, water content and compactive effort on the compaction of selected California forest and range soils. *Soil Sci. Soc. Am. J.* 45: 231–236.
- Howard, W. E. 1964. Introduced browsing mammals and habitat stability in New Zealand. *J. Wildlife Manage.* 28: 421–429.
- Huang, J., Lacey, S. T. & Ryan, P. J. 1996. Impact of forest harvesting on the hydraulic properties of surface soil. *Soil Sci.* 161: 79–86.
- Hullugalle, N. R. & Lal, R. 1986. Root growth of maize in a compacted gravelly tropical alisol as affected by rotation with a woody perennial. *Field Crops Res.* 13: 33–44.

- Humphreys, F. R. & Craig, F. G. 1981. Effects of fire on soil chemical, structural and hydrological processes. *In* Gill, A.M. & Graves, R.H. (eds.). *Fire and the Australian Biota*, pp. 177–200. Griffin Press, Netley, South Australia.
- Jim, C. Y. 1987a. Camping impacts on vegetation and soil in a Hong Kong country park. *Appl. Geog.* 7: 317–332.
- Jim, C. Y. 1987b. Trampling impacts of recreationists on picnic sites in a Hong Kong country park. *Environ. Conserv.* 14: 117–127.
- Jim, C. Y. 1993. Soil compaction as a constraint to tree growth in tropical and subtropical urban habitats. *Environ. Conserv.* 20: 35–49.
- Johnson, C. E., Johnson, A. H., Huntington, T. G. & Siccama, T. G. 1991. Whole-tree clear-cutting effects on soil horizons and organic matter pools. *Soil Sci. Soc. Am. J.* 55: 497–502.
- Joshi, O. P., Manikandan, P. & Khan, H. H. 1987. Effect of soil compaction on infiltration capacity of a red sandy loam soil arenic paleustult. *J. Ind. Soc. Soil Sci.* 35: 286–289.
- Jusoff, K. 1989. Physical soil properties associated with recreational use of a forested reserve area in Malaysia. *Environ. Conserv.* 16: 339–342.
- Kamaruzaman, J. 1991. Effect of tracked and rubber-tired logging machines on soil physical properties of the Berkelah Forest Reserve, Malaysia. *Pertanika* 14: 265–276.
- Kamaruzaman, J. & Nik Muhamad, M. 1992a. An analysis of soil disturbance from logging operation in a hill forest of Peninsular Malaysia. *For. Ecol. Manage.* 47: 323–333.
- Kamaruzaman, J. & Nik Muhamad, M. 1992b. Effect of crawler tractor logging on soil compaction in central Pahang, Malaysia. *Malays. Forester* 50: 274–280.
- Kang, B. T. & Lal, R. 1981. Nutrient losses in water runoff from agricultural catchments. *In* Lal, R. & Russel, E. W. (eds.). *Tropical Agricultural Hydrology*, pp. 153–161. Wiley, New York.
- Kays, B. L. & Patterson, J. C. 1982. Soil drainage and infiltration. *In* Craul, P. J. (editor). *Urban Forest Soils: A Reference Workbook*, pp. 5.1–5.25. College of Environmental Sci. and Forestry, Syracuse, NY.
- Kays, S. J., Nicklow, C. W. & Simons, D. W. 1974. Ethylene in relation to the response of roots to physical impedance. *Plant Soil* 40: 565–571.
- Kesik, T. & Stanek, R. 1983. Reakcja krzewow porzeczek i malin na zageszczenie gleby. *Agrotechnika sadow ze szczegolnym uwzglednieniem terenow erodowanych*, Lublin, Poland, pp. 41–44 (in Polish).
- Khaleel, R., Reddy, K. R. & Overcash, M. R. 1981. Changes in soil physical properties due to organic waste application. *J. Environ. Qual.* 10: 133–141.
- Kliejunas, J. T. 1992. A biological evaluation of soil tilling and the potential for pest problems with changing management practices on the Milford Ranger District, Plumas National Forest. USDA Forest Service, Pacific Southwest Region, Forest Pest Management Report no. R92-09, 13 pp.
- Knapp, P. A. 1989. Natural recovery of compacted soil in semi-arid Montana. *Phys. Geog.* 10: 165–175.
- Koolen, A. J. & Kuipers, H. 1983. *Agricultural Soil Mechanics*. Springer, Berlin.
- Kopinga, J. 1985a. Site preparation practices in the Netherlands. *In* Kuhns, L. J. & Patterson, L. C. (eds.). *Selecting and Preparing Sites for Urban Trees*, pp. 62–71. Pennsylvania State University, University Park, PA.
- Kopinga, J. 1985b. Research on street tree-planting practices in the Netherlands. *In* Kuhns, L. J. & Patterson, L. C. (eds.). *Selecting and Preparing Sites for Urban Trees*, pp. 72–84. Pennsylvania State University, University Park, PA.
- Korotaev, A. A. 1992. Bodenverdichtung und Wurzelwachstum der Bäume. *Forstarchiv* 63: 116–119.
- Kozłowski, T. T. 1972a. Shrinking and swelling of plant tissues. *In* Kozłowski, T.T. (editor). *Water Deficits and Plant Growth*, Vol. III, pp.1–64. Academic Press, New York.
- Kozłowski, T. T. 1972b. Physiology of water stress. *In* Wildland Shrubs—Their Biology and Utilization, pp. 229–244. US Forest Service General Technical Report no. INT-1.
- Kozłowski, T. T. 1976. Drought resistance and transplantability of shade trees. *In* Santamour, F. S., Gerhold, H. & Little, S. (editors). *Better Trees for Metropolitan Landscapes*, pp. 77–90. USDA Forest Service General Technical Report NE-22 Upper Darby, PA.
- Kozłowski, T. T. 1982a. Water supply and tree growth. Part I. Water deficits. *For. Abstr.* 43: 57–95.
- Kozłowski, T. T. 1982b. Water supply and tree growth. Part II. Flooding. *For. Abstr.* 43: 145–161.
- Kozłowski, T. T. 1984a. Extent, causes, and impacts of flooding. *In* Kozłowski, T.T. (ed.). *Flooding and Plant Growth*, pp. 1–7. Academic Press, Orlando, FL.
- Kozłowski, T. T. 1984b. Responses of woody plants to flooding. *In* Kozłowski, T.T. (ed.). *Flooding and Plant Growth*, pp. 129–163. Academic Press, Orlando, FL.
- Kozłowski, T. T. 1984c. Plant responses to flooding of soil. *BioScience* 34: 162–167.
- Kozłowski, T. T. 1985. Soil aeration, flooding, and tree growth. *J. Arboricult.* 11 (3): 85–96.
- Kozłowski, T. T. 1986. Soil aeration and growth of forest trees (Review article). *Scand. J. For. Res.* 1: 113–123.
- Kozłowski, T. T. 1997. Responses of woody plants to flooding and salinity. Online. World wide web at <http://heronpublishing.com/tp/monograph/kozłowski.pdf>. 29 pp.
- Kozłowski, T. T. & Pallardy, S. G. 1997a. *Physiology of Woody Plants*, 2nd edn. Academic Press, San Diego, CA.
- Kozłowski, T. T. & Pallardy, S. G. 1997b. *Growth Control in Woody Plants*. Academic Press, San Diego, CA.
- Kozłowski, T. T., Kramer, P. J. & Pallardy, S. G. 1991. *The Physiological Ecology of Woody Plants*. Academic Press, San Diego, CA.
- Kramer, P. J. & Boyer, J. S. 1995. *Water Relations of Plants and Soils*. Academic Press, San Diego, CA.

- Krenzer, E. G., Jr., Chee, C. F. & Stone, J. F. 1989. Effects of animal traffic on soil compaction in wheat pastures. *J. Prod. Agric.* 2: 246–249.
- Kubiske, M. E. & Abrams, M. D. 1993. Stomatal and nonstomatal limitations of photosynthesis in 19 temperate tree species on contrasting sites during wet and dry years. *Plant. Cell Environ.* 16: 1123–1129.
- Lachno, D. R., Harrison-Murray, R. S. & Audus, L. J. 1982. The effects of mechanical impedance to growth on the levels of ABA and IAA in root tips of *Zea mays*. *J. Exp. Bot.* 33: 943–951.
- Lakso, A. N. 1979. Seasonal changes in stomatal response to leaf water potential in apple. *J. Am. Soc. Hortic. Sci.* 104: 58–60.
- Larson, W. E., Eynard, A., Hadas, A. & Lipiec, J. 1994. Control and avoidance of soil compaction in practice. In Soane, B. J. & van Ouwerkerk, C. (eds.). *Soil Compaction and Crop Production*, pp. 597–625. Elsevier, Amsterdam.
- Legg, M. H. & Schneider, G. 1977. Soil deterioration in campsites: northern forest types. *Soil Sci. Soc. Am. Proc.* 41: 437–441.
- Lenhard, R. J. 1986. Changes in void distribution and volume during compaction of a forest soil. *Soil Sci. Soc. Am. J.* 50: 462–464.
- Liang, J., Zhang, J. & Wong, M. H. 1996. Stomatal conductance in relation to abscisic acid concentrations in two tropical trees, *Acacia confusa* and *Litsea glutinosa*. *Plant Cell Environ.* 19: 93–100.
- Linn, D. M. & Doran, J. W. 1984. Aerobic and anaerobic microbial populations in no-till and plowed soils. *Soil Sci. Soc. Am. J.* 48: 794–799.
- Lockaby, B. G. & Vidrine, C. G. 1984. Effect of logging equipment traffic on soil density and growth and survival of loblolly pine. *South. J. Appl. For.* 8: 109–112.
- Lodge, D. J. 1986. Effects of soil moisture, drainage, and microbial interactions on formation of endo- and ectomycorrhizae in eastern cottonwood. *Phytopathology* 76: 1110.
- Lowery, B. & Schuler, R. T. 1991. Temporal effects of subsoil compaction on soil strength and plant growth. *Soil Sci. Soc. Am. J.* 55: 216–223.
- Lowry, F. E., Taylor, H. M. & Huck, M. G. 1970. Growth rate and yield of cotton as influenced by depth and bulk density of soil pans. *Soil Sci. Soc. Am. Proc.* 34: 306–309.
- Lunt, O. R. 1956. Minimizing compaction in putting greens. *USDA J. Turf Manage.* 9 (5): 25–30.
- Lutz, H. 1945. Soil conditions on picnic grounds in public forest parks. *J. For.* 43: 121–127.
- McAfee, M., Lindstrom, J. & Johnson, W. 1989. Effects of pre-sowing compaction on soil physical properties, soil atmosphere and growth of oats on a clay soil. *J. Soil Sci.* 40: 707–717.
- McColl, J. G. 1995. Forest clear-cutting, soil response. In Nierenberg, W. A. (ed.). *Encyclopedia of Environmental Biology*, Vol. 2, pp. 95–103. Academic Press, San Diego, CA.
- McConnell, J. S., Frizzell, B. S. & Wilkerson, M. H. 1989. Effects of soil compaction and subsoil tillage of two alfisols on the growth and yield of cotton. *J. Prod. Agric.* 2: 140–146.
- McCoy, E. L. 1992. Quantitative physical assessment of organic materials used in sports turf root zone mixes. *Agron. J.* 84: 375–381.
- McNeel, J. F. & Ballard, T. M. 1992. Analysis of site stand impacts from thinning with a harvester-forwarder system. *J. For. Engin.* 4: 23–29.
- MacFall, J. S., Slack, S. A. & Iyer, J. 1991. Effects of *Hebeloma arenosa* and phosphorus fertility on growth of red pine (*Pinus resinosa*) seedlings. *Can. J. Bot.* 69: 372–379.
- Madison, J. H. 1971. *Practical Turfgrass Management*. Van-Nostrand-Reinhold, New York.
- Malmer, A. & Grip, H. 1990. Soil disturbance and loss of infiltrability caused by mechanized and manual extraction of tropical rainforest in Sabah, Malaysia. *For. Ecol. Manage.* 38: 1–12.
- Marschner, H. 1995. *Mineral Nutrition of Plants*, 2nd edn. Academic Press, London.
- Marsh, C. W., Tay, J., Pinard, M. A., Putz, F. E. & Sullivan, T. E. 1996. Reduced impact logging in Sabah, Malaysia. In Schulte, A. & Schöne, D. (eds.). *Dipterocarp Forest Ecosystems*, pp. 293–307. World Scientific, Singapore.
- Masle, J. & Passioura, J. B. 1987. The effect of soil strength on the growth of young wheat plants. *Aust. J. Plant Physiol.* 14: 643–656.
- Mason, E. G. & Cullen, A. W. J. 1986. Growth of *Pinus radiata* on ripped and unripped taupo pumice soil. *N Z J. For. Sci.* 16: 3–18.
- Materechera, S. A., Alston, A. M., Kirby, J. M. & Dexter, A. R. 1992. Influence of root diameter on the penetration of seminal roots into a compacted subsoil. *Plant Soil* 144: 297–303.
- Miller, R. C., Scott, W. & Hazard, J. W. 1996. Soil compaction and conifer growth after tractor yarding at three coastal Washington locations. *Can. J. For. Res.* 26: 225–236.
- Minore, D., Smith, C. E. & Woolard, R. F. 1969. Effects of high soil density on seedling root growth of seven northwestern tree species. *USDA Forest Service Research Note PNW-112*, 6 pp.
- Misra, R. K. & Gibbons, A. K. 1996. Growth and morphology of eucalypt seedling roots in relation to soil strength arising from compaction. *Plant Soil* 182: 1–11.
- Mitchell, M. L., Hassan, A. E., Davey, C. B. & Gregory, J. D. 1982. Loblolly pine growth in compacted greenhouse soils. *Trans. Am. Soc. Agric. Eng.* 25: 304–307, 312.
- Mohd. Basri, H. & Nik Muhamad, M. 1992. Effects of soil compaction on growth of three-year old *Acacia mangium* Willd. stand in Setul, Negri Sembilan, Malaysia. *Malays. Forester* 50: 250–257.
- Monti, P. W. & Mackintosh, E. E. 1979. Effects of camping on surface soil properties in the boreal forest region of northwestern Ontario, Canada. *Soil Sci. Soc. Am. J.* 43: 1024–1029.
- Mulholland, B. J. & Fullen, M. A. 1991. Cattle trampling and soil compaction on loamy sands. *Soil Use Manage.* 7: 189–193.

- Mulholland, B. J., Taylor, I. B., Roberts, J. A. & Black, C. R. 1995. Soil compaction. ABA and root-to-shoot signalling. *J. Exp. Bot.* 46 (Suppl.): 26.
- Mulholland, B. J., Black, C. R., Taylor, I. B., Roberts, J. A. & Lenton, J. R. 1996. Effect of soil compaction on barley (*Hordeum vulgare*) growth. I. A possible role for ABA as a root sourced chemical signal. *J. Exp. Bot.* 47: 539–549.
- Mullins, C. E. 1991. Physical properties of soils in urban environments. In Bullock, P. & Gregory, P. J. (eds.). *Soils in the Urban Environment*, pp. 87–118. Blackwell, London.
- Murphy, G. 1983. *Pinus radiata* survival, growth and form four years after planting off and on skidtrails. *N Z J. For.* 28: 184–193.
- Murphy, J. A., Rieke, P. E. & Erickson, A. E. 1993. Core cultivation of a putting green with hollow and solid tines. *Agron. J.* 85: 1–9.
- Nambiar, E. K. S. & Sands, R. 1992. Effects of compaction and simulated root channels in the subsoil on root development, water uptake and growth of radiata pine. *Tree Physiol.* 10: 297–306.
- Ni, B. R. & Pallardy, S. G. 1992. Stomatal and nonstomatal limitations to net photosynthesis in seedlings of woody angiosperms. *Plant Physiol.* 99: 1502–1508.
- Nimmo, J. R. & Akstin, K. C. 1988. Hydraulic conductivity of a sandy soil at low water content after compaction by various methods. *Soil Sci. Soc. Am. J.* 52: 303–310.
- Nommik, H. & Larsson, K. 1989. Measurement of denitrification rate in undisturbed soil areas under different temperature and moisture conditions using nitrogen¹⁵ tracer technique: 2. Factors affecting denitrification. *Swed. J. Agric. Res.* 19: 35–44.
- Nortcliff, S., Ross, S. M. & Thornes, J. B. 1990. Soil moisture, runoff and sediment yield from differentially cleared tropical rainforest plots. In Thomas, J. B. (ed.). *Vegetation and Erosion: Processes and Environment*, pp. 419–436. Wiley, Chichester.
- Nussbaum, R. & Hoe, A. L. 1990. Rehabilitation of degraded sites in logged-over forests using dipterocarps. In Schulte, A. & Schöne, D. (eds.). *Dipterocarp Forest Ecosystems*, pp. 446–463. World Scientific, London.
- O'Keefe, D. M. & Sylvia, D. M. 1991. Mechanisms of the vesicular-arbuscular plant growth response. In Arora, D. K., et al. (eds.). *Handbook of Applied Mycology*, pp. 35–54. Marcel Dekker, New York.
- Ole-Meiludie, R. E. L. & Njau, W. L. M. 1989. Impact of logging equipment on water infiltration capacity at Olmotonyi, Tanzania. *For. Ecol. Manage.* 26: 207–213.
- Osborne, D. J. 1976. Control of cell shape and cell size by the dual regulation of auxin and ethylene. In Sunderland, N. (ed.). *Perspectives in Experimental Botany*, pp. 89–102. Pergamon Press, Oxford.
- Otrosina, W. J., Sung, S.-J. & White, L. M. 1996. Effects of subsoiling on lateral roots, sucrose metabolizing enzymes, and soil ergosterol in two Jeffrey pine stands. *Tree Physiol.* 16: 1009–1013.
- Pan, E. & Bassuk, N. 1985. Effects of soil type and compaction on the growth of *Ailanthus altissima* seedlings. *J. Environ. Horticult.* 3: 158–162.
- Patterson, J. C. 1975. Enrichment of urban soil with composted sewage sludge and leaf mold—Constitution Gardens. *Compost Sci.* 16 (3): 18–21.
- Patterson, J. C. 1976. Soil compaction and its effects upon urban vegetation. In Santamour, F. S., Gerhold, H. D. & Little, S. (editors). *Better Trees for Metropolitan Landscapes*, pp. 91–101. USDA Forest Service General Technical Report NE-22, Upper Darby, PA.
- Patterson, J. C. 1977. Soil compaction—effects on urban vegetation. *J. Arboricult.* 3: 161–167.
- Patterson, J. C. & Mader, D. L. 1982. Soil compaction: causes and control. In Craul, P. J. (editor). *Urban Forest Soils: A Reference Workbook*, pp. 3.1–3.15. College of Environmental Science and Forestry, Syracuse, NY.
- Pereira, J. S. & Kozłowski, T. T. 1977. Variations among woody angiosperms in response to flooding. *Physiol. Plant.* 41: 184–192.
- Pezeshki, S. R. & Chambers, J. L. 1985. Responses of cherrybark oak (*Quercus falcata* var. *pagodaefolia*) seedlings to short-term flooding. *For. Sci.* 31: 760–771.
- Pinard, M., Howlett, B. & Davidson, D. 1996. Site conditions limit pioneer tree recruitment after logging of dipterocarp forests in Sabah, Malaysia. *Biotropica* 28: 2–12.
- Pittenger, D. R. & Stamen, T. 1990. Effectiveness of methods used to reduce harmful effects of compacted soil around landscape trees. *J. Arboricult.* 16 (3): 55–57.
- Plass, M. C. & Van der Bruijnzeel, L. A. 1993. Impact of mechanized selective logging on topsoil infiltrability in the upper Segoma area, Sabah, Malaysia. In Gladwell, J. S. (ed.). *Hydrology of Warm Humid Regions*, pp. 203–211. IAHS Press, Wallingford.
- Pomeroy, K. B. 1949. The germination and initial establishment of loblolly pine under various surface soil conditions. *J. For.* 47: 541–543.
- Ponnamperuma, F. N. 1984. Effects of flooding on soils. In Kozłowski, T. T. (ed.). *Flooding and Plant Growth*, pp. 9–45. Academic Press, Orlando, FL.
- Powlson, D. S. 1993. Understanding the nitrogen cycle. *Soil Use Manage.* 9: 86–94.
- Purser, M. D. & Cundy, T. W. 1992. Changes in soil physical properties due to cable yarding and their hydrologic implications. *West. J. Appl. For.* 7: 36–39.
- Rab, M. A. 1994. Changes in physical properties of a soil associated with logging of *Eucalyptus regnans* forest in southwestern Australia. *For. Ecol. Manage.* 70: 215–229.
- Randrup, T. B. 1997. Soil compaction on construction sites. *J. Arboricult.* 23: 207–210.
- Regehr, D. L., Bazzaz, F. A. & Boggess, W. R. 1975. Photosynthesis, transpiration and leaf conductance of *Populus deltoides* in relation to flooding and drought. *Photosynthetica* 9: 52–61.
- Reisinger, T. W., Simmons, G. L. & Pope, P. E. 1988. The impact of timber harvesting on soil properties, and seedling growth in the south. *South. J. Appl. For.* 12: 58–67.
- Richards, D. & Cockroft, B. 1974. Soil physical properties and root concentrations in an irrigated apple orchard. *Aust. J. Exp. Agric. Anim. Husb.* 14: 103–107.

- Rolf, R. K. 1991. Soil improvement and increased growth response from subsoil cultivation. *J. Arboricult.* 17: 200–201.
- Rolf, R. K. 1992. Soil physical effects of pneumatic subsoil loosening using a Terralift soil aerator. *J. Arboricult.* 18: 235–248.
- Rolf, R. K. 1994. A review of preventative and loosening measures to alleviate soil compaction in tree planting areas. *Arboricult. J.* 18: 431–448.
- Rosen, C. J. & Carlson, P. M. 1984. Influence of root zone oxygen stress on potassium and ammonium absorption by Myrobalan plum rootstock. *Plant Soil* 80: 345–353.
- Roth, C. H. & Joschko, M. 1991. A note on the reduction of runoff from crusted soil by earthworm burrows and artificial channels. *Pflanzen Bodenkunde* 154: 101–106.
- Ruark, G. A., Mader, D. L. & Tattar, T. A. 1982. The influence of soil compaction and aeration on the root growth and vigor of trees—a literature review. Part I. *Arboricult. J.* 6: 251–265.
- Ruark, G. A., Mader, D. L. & Tattar, T. A. 1983. The influence of soil moisture and temperature on the root growth and vigor of trees—a literature review. Part II. *Arboricult. J.* 7: 39–51.
- Sands, R. & Bowen, G. D. 1978. Compaction of sandy soils in radiata pine forests. II. Effects of compaction on root configuration and growth of radiata pine seedlings. *Aust. For. Res.* 8: 163–170.
- Sarquis, J. I., Jordan, W. R. & Morgan, P. W. 1991. Ethylene evolution from maize (*Zea mays* L.) seedling roots and shoots in response to mechanical impedance. *Plant Physiol.* 96: 1171–1177.
- Sarquis, J. I., Morgan, P. W. & Jordan, W. R. 1992. Metabolism of 1-aminocyclopropane-1-carboxylic acid in etiolated maize seedlings grown under mechanical impedance. *Plant Physiol.* 98: 1342–1348.
- Settergren, C. D. & Cole, D. M. 1970. Recreation effects on soil and vegetation in the Missouri Ozarks. *J. For.* 68: 231–233.
- Sharda, A. K. 1977. Influence of soil bulk density on horizontal water infiltration. *Aust. J. Soil Res.* 15: 83–86.
- Sheriff, D. W. & Nambiar, E. K. S. 1995. Effect of subsoil compaction and three densities of simulated root channels in the subsoil on growth, carbon gain and water uptake of *Pinus radiata*. *Aust. J. Plant Physiol.* 22: 1001–1013.
- Shetron, S. G., Sturos, J. A., Padley, E. & Trettini, C. 1988. Forest soil compaction: effect of multiple passes and loadings on wheel track surface soil bulk density. *North. J. Appl. For.* 5: 120–123.
- Short, J. R., Fanning, D. S., Foss, J. E. & Patterson, J. C. 1986. Soils of the mall in Washington, D.C. II. Genesis, classification and mapping. *Soil Sci. Soc. Am. J.* 50: 705–710.
- Shoulders, E. & Ralston, C. W. 1975. Temperature, root aeration and light influence slash pine nutrient uptake rates. *For. Sci.* 21: 401–410.
- Shuja, N., Gilmi, U. & Kahn, A. G. 1971. Mycorrhizal associations in some angiosperm trees around New University Campus, Lahore. *Pak. J. For.* 21: 367–373.
- Silberbush, M., Hallmark, W. B. & Barber, S. A. 1983. Simulation of effects of soil bulk density and P addition on K uptake by soybeans. *Commun. Soil Sci. Plant Anal.* 14: 287–296.
- Sills, M. J. & Carrow, R. N. 1983. Turfgrass growth, N use, and water use under soil compaction and N fertilization. *Agron. J.* 75: 488–492.
- Simmons, G. L. & Pope, P. E. 1987. Influence of soil compaction and vesicular-arbuscular mycorrhizae on root growth of yellow poplar and sweet gum seedlings. *Can. J. For. Res.* 17: 970–975.
- Simmons, G. L. & Pope, P. E. 1988a. Using VA-mycorrhizae to enhance seedling root growth in compacted soil. *North. J. Appl. For.* 5: 65–68.
- Simmons, G. L. & Pope, P. E. 1988b. Influence of soil water potential and mycorrhizal colonization on root growth of yellow-poplar and sweet gum seedlings grown in compacted soil. *Can. J. For. Res.* 18: 1392–1396.
- Smith, D. L. O. 1987. Measurement, interpretation and modelling of soil compaction. *Soil Use Manage.* 3: 87–93.
- Snider, M. D. & Miller, R. F. 1985. Effects of tractor logging on soils and vegetation in eastern Oregon. *Soil Sci. Soc. Am. J.* 49: 1280–1282.
- Soane, B. D. 1990. The role of organic matter in soil compactibility: a review of some practical aspects. *Soil Tillage Res.* 16: 179–202.
- Soane, B. D., Blackwell, P. S., Dickson, J. W. & Painter, D. J. 1981a. Compaction by agricultural vehicles: a review. I. Soil and wheel characteristics. *Soil Tillage Res.* 1: 207–237.
- Soane, B. D., Blackwell, P. S., Dickson, J. W. & Painter, D. J. 1981b. Compaction by agricultural vehicles: a review. II. Compaction under tyres and other running gear. *Soil Tillage Res.* 1: 373–400.
- Soane, B. D., Dickson, J. W. & Campbell, D. J. 1982. Compaction by agricultural vehicles: a review. III. Incidence and control of compaction in crop production. *Soil Tillage Res.* 2: 3–36.
- Sojka, R. E., Westermann, D. T., Brown, M. J. & Meek, B. D. 1993. Zone-subsoiling effects on infiltration, runoff, erosion, and yields of furrow-irrigated potatoes. *Soil Tillage Res.* 25: 351–368.
- Spoor, G. 1980. Subsoiling and deep cultivation. *J. Sci. Fd Agric.* 31: 418–419.
- Springett, J. A., Gray, R. A. J. & Reid, J. B. 1992. Effect of introducing earthworms into horticultural land previously denuded of earthworms. *Soil Biol. Biochem.* 16: 1615–1622.
- Steinbrenner, E. C. 1955. The effect of repeated tractor trips on the physical properties of forest soils. *Northwest Sci.* 29: 155–159.
- Steinbrenner, E. C. & Gessell, S. P. 1955. The effects of tractor logging on physical properties of some forest soils in southwestern Washington. *Soil Sci. Soc. Am. Proc.* 19: 372–376.
- Stepniewski, W., Glinski, J. & Ball, B. C. 1994. Effects of compaction on soil aeration properties. In Soane, B. D. & van Ouwerkerk, C. (eds.). *Soil Compaction and Crop Production*, pp. 167–189. Elsevier, Amsterdam.

- Stewart, R., Froehlich, H. & Olsen, E. 1988. Soil compaction: an economic model. *West. J. Appl. For.* 3: 20–22.
- Stirzaker, R. J., Passioura, J. B. & Wilms, Y. 1996. Soil structure and plant growth: impact of bulk density and biopores. *Plant Soil* 185: 151–162.
- Stolzy, L. H. & Sojka, R. E. 1984. Effects of flooding in plant disease. In Kozłowski, T. T. (ed.). *Flooding and Plant Growth*, pp. 261–264. Academic Press, New York.
- Stolzy, L. H., Letey, J., Klotz, L. J. & Labanauskas, C. K. 1965. Water and aeration as factors in root decay of *Citrus sinensis*. *Phytopathology* 55: 270–275.
- Tackett, J. L. & Pearson, R. W. 1964. Oxygen requirements for cotton seedling root penetration of compacted soil cores. *Soil Sci. Soc. Am. Proc.* 28: 600–605.
- Tang, Z. C. & Kozłowski, T. T. 1982. Some physiological and morphological responses of *Quercus macrocarpa* seedlings to flooding. *Can. J. For. Res.* 10: 308–311.
- Tanner, C. B. & Mamaril, C. P. 1959. Pasture soil compaction by animal traffic. *Agron. J.* 51: 329–331.
- Tardieu, F. 1994. Growth and functioning of roots and of root systems subjected to soil compaction: towards a system with multiple signalling. *Soil Tillage Res.* 30: 217–243.
- Tardieu, F., Katerji, N., Bethenod, O., Zhang, J. & Davies, W. J. 1991. Maize stomatal conductance in the field: its relationship with soil and plant water potential, mechanical constraints and ABA concentration in the xylem sap. *Plant Cell Environ.* 14: 121–126.
- Tardieu, F., Zhang, J., Katerji, N., Bethenod, O., Palmer, S. & Davies, W. J. 1992. Xylem ABA controls the stomatal conductance of field-grown maize subjected to soil compaction or soil drying. *Plant Cell Environ.* 15: 193–197.
- Taylor, H. M., Robertson, G. M. & Parker, J. J. 1966. Soil strength–root penetration relations for medium to coarse-textured soil materials. *Soil Sci.* 102: 18–22.
- Tenhunen, J. D., Lange, O. L. & Jahner, D. 1982. The control by atmospheric factors and water stress of mid-day stomatal closure in *Arbutus unedo* growing in a natural macchia. *Oecologia* 55: 165–169.
- Teskey, R. O., Fites, J. A., Samuelson, L. J. & Bongarton, B. C. 1986. Stomatal and non-stomatal limitations to net photosynthesis in *Pinus taeda* L. under different environmental conditions. *Tree Physiol.* 2: 131–142.
- Thithamer, J. 1989. Über die Einflüsse der Bodendichte und des Humusgehalts auf die Wuchsleistungen von Kiefern und Douglasien sandiger Boden in Forstbaumschulen. *Beiträge Forstwirtschaft.* 23: 25–29.
- Thorud, D. B. & Frissell, S. S. 1976. Time changes in soil density following compaction under an oak forest. *Minnesota Forestry Research Note* 257.
- Tiedje, J. M. 1988. Ecology of denitrification and dissimilatory nitrate reduction to ammonium. In Zehnder, A. J. B. (ed.). *Biology of Anaerobic Organisms*, pp. 179–246. Wiley, New York.
- Torbert, H. A. & Wood, C. W. 1992. Effects of soil compaction and water-filled porespace on soil microbial activity and N losses. *Commun. Soil Sci. Plant Anal.* 23: 1321–1331.
- Trouse, A. C., Jr., 1983. Observations on under-the-row subsoiling after conventional tillage. *Soil Tillage Res.* 3: 67–81.
- Tuttle, C. L., Golden, M. S. & Meldahl, R. S. 1988. Soil compaction effects on *Pinus taeda* establishment from seed and early growth. *Can. J. For. Res.* 18: 628–632.
- Unger, P. W. & Kaspar, T. C. 1994. Soil compaction and root growth: a review. *Agron. J.* 86: 759–766.
- Usman, H. 1994. Cattle trampling and soil compaction effects on soils properties of the northeastern Nigerian sandy loam. *Arid Soil Res. Rehab.* 8: 69–75.
- Van Noordwijk, M., Widiyanto, H. M. & Hairiah, K. 1991. Old tree root channels in acid soils in the humid tropics: important for crop production, water infiltration and nitrogen management. *Plant Soil* 134: 37–44.
- Vepraskas, M. J. 1994. Plant response mechanisms to soil compaction. In Wilkinson, R. E. (ed.). *Plant–Environment Interactions*, pp. 263–287. Marcel Dekker, New York.
- Vermes, J.-F. & Myrold, D. D. 1992. Denitrification in forest soils of Oregon. *Can. J. For. Res.* 22: 504–512.
- Voorhees, W. B. 1983. Relative effectiveness of tillage and natural forces in alleviating wheel-induced soil compaction. *Soil Sci. Soc. Am. J.* 47: 129–133.
- Vora, R. S. 1988. Potential soil compaction forty years after logging in northeastern California. *Great Basin Nat.* 48: 117–120.
- Wairiu, M., Mullins, C. E. & Campbell, C. D. 1993. Soil physical factors affecting the growth of sycamore (*Acer pseudoplatanus* L.) in a silvopastoral system on a stony upland soil in north-east Scotland. *Agrofor. Syst.* 24: 295–306.
- Wang, J., Hesketh, J. D. & Wooley, J. T. 1986. Preexisting channels and soybean rooting patterns. *Soil Sci.* 141: 432–437.
- Warren, S. D., Thurow, T. L., Blackburn, W. H. & Garza, N. E. 1987. The influence of livestock trampling under intensive rotation grazing on soil hydrologic characteristics. *J. Range Manage.* 39: 491–495.
- Wästerlund, I. 1985. Compaction of till soils and growth tests with Norway spruce and Scots pine. *For. Ecol. Manage.* 11: 171–179.
- Wästerlund, I. 1988. Damages and growth effects after selective mechanical clearing. *Scand. J. For. Res.* 3: 259–272.
- Webb, R. H. & Stielstra, S. S. 1979. Sheep grazing effects on Mojave Desert vegetation and soils. *Environ. Manage.* 3: 517–529.
- van der Weert, R. 1974. The influence of mechanical forest clearing on soil conditions and resulting effects on root growth. *Trop. Agric.* 51: 325–331.
- Wells, C. G., Campbell, R. E. & DeBano, L. F. 1979. Effects of fire on soil. A state of knowledge review. USDA General Technical Report WO-7. Washington, DC.
- Wert, S. & Thomas, B. R. 1981. Effects of skid roads on diameter, height and volume growth of Douglas fir. *Soil Sci. Soc. Am. J.* 45: 629–632.

- Weyers, M., Schroeder, D. & Martinovic, L. 1993. Studies for melioration and tillage of compacted recultivated soils in Reinisches Braunkohlenggebiet Rhenish Brown Coal District. *Zeitschr. Kulturtech. Landentwick.* 34: 165–174.
- Whiteley, G. M. & Dexter, A. R. 1983. Behaviour of roots in cracks between soil peds. *Plant Soil* 74: 153–162.
- Whitman, A. A., Brokaw, N. V. L. & Hagan, J. M. 1997. Forest damage caused by logging of mahogany (*Swietenia macrophylla*) in northern Belize. *For. Ecol. Manage.* 92: 87–96.
- Wieniarska, J., Lipecki, J., Stanek, R. & Kesik, T. 1987. The effects of soil compaction due to machinery operation on a raspberry plantation. *Fruit Sci. Rep. (Skierniewice)* 14 (2): 71–78.
- Wilde, S. A. 1954. Mycorrhizal fungi: their distribution and effect on tree growth. *Soil Sci.* 78: 23–31.
- Willatt, S. T. & Pullar, D. M. 1983. Changes in soil physical properties under grazed pastures. *Aust. J. Soil Res.* 22: 343–348.
- Wingate-Hill, R. & Jakobson, B. F. 1982. Increased mechanisation and soil damage in forests: a review. *N Z J. For. Sci.* 12: 380–393.
- Wolkowski, R. P. 1990. Relationship between wheel-traffic-induced soil compaction, nutrient availability and crop growth: a review. *J. Prod. Agric.* 3: 460–469.
- Wood, J. C., Blackburn, W. H., Pearson, H. A. & Hunter, T. K. 1989. Infiltration and runoff water quality response to silvicultural and grazing treatments on a longleaf pine forest. *J. Range Manage.* 42: 378–381.
- Woodward, C. L. 1996. Soil compaction and topsoil removal effects on soil properties and seedling growth in Amazonian Ecuador. *For. Ecol. Manage.* 82: 197–209.
- Wright, H. A. & Bailey, A. W. 1982. *Fire Ecology*. Wiley, New York.
- Wronski, E. B. & Murphy, G. 1994. Responses of forest crops to soil compaction. In Soane, B. D. & van Ouwerkerk, C. (eds.). *Soil Compaction in Crop Production*, pp. 317–342. Elsevier, New York.
- Yelenosky, G. 1963. Soil aeration and tree growth. *Proc. Int. Shade Tree Conf.* 39: 16–25.
- Yingling, E. L., Keeley, C. A., Little, S. & Brutis, G., Jr., 1979. Reducing damage to shade and woodland trees from construction activities. *J. Arboricult.* 5: 97–104.
- Young, I. A. & Bengough, A. G. 1989. Mechanical constraints to root growth. In *Roots and the Soil Environment*, pp. 41–47. Aspects of Applied Biology, Wellesbourne. Association of Applied Biologists, Institute of Horticultural Research, Warwick, Scotland.
- Young, R. A. & Gilmore, A. R. 1976. Effects of various camping intensities on soil properties in Illinois campgrounds. *Soil Sci. Soc. Am. J.* 40: 908–911.
- Youngberg, C. T. 1959. The influence of soil conditions following tractor logging on the growth of planted Douglas-fir seedlings. *Soil Sci. Soc. Am. Proc.* 23: 76–78.
- Zabowski, D., Skinner, M. F. & Rygiewicz, P. T. 1994. Timber harvesting and long-term productivity: weathering processes and soil disturbance. *For. Ecol. Manage.* 66: 55–68.
- Zander, J., Ammer, U. & Breitsameter, J. 1988. Experimentelle Untersuchungen über den Einfluss des Befahrens auf die Oberbodenverdichtung. *Forstwissen Centralblatt* 107: 112–122.
- Zhang, J. & Davies, W. J. 1990. Changes in the concentration of ABA in xylem sap as a function of changing soil water status can account for changes in leaf conductance and growth. *Plant Cell Environ.* 13: 277–286.
- Zisa, R. P., Halverson, H. G. & Stout, B. J. 1980. Establishment and early growth of conifers on compact soils in urban areas. USDA Forest Service Research Paper NE-451.